

Advanced Signal Processing

(Music and Speech Signal Processing)

1st lecture (4/07)

Graduate School of Information Science and Technology
Creative Informatics · Information Physics & Computing

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Advanced Signal Processing (2026 year)

- 04/07: 1st Statistical speech & acoustic signal Processing
- 04/14: 2nd Nonnegative matrix factorization
- 04/21: 3rd Blind source separation 1
- 04/28: 4th Blind source separation 2
- 05/12: 5th Enhancement & higher-order statistics
- 05/19: 6th 【Report-1】
- 05/26: 7th Multi-channel acoustic signal processing
- 06/09: 8th Music information processing
- 06/16: 9th Environmental sound generation & recognition
- 06/23: 10th Speech synthesis & voice conversion
- 06/30: 11th 【Report-2】

Lecture materials and performance evaluation

- Lecture materials
 - On UTOL (ex-ITC-LMS)
- Performance evaluation
 - By reports (two submissions are required)

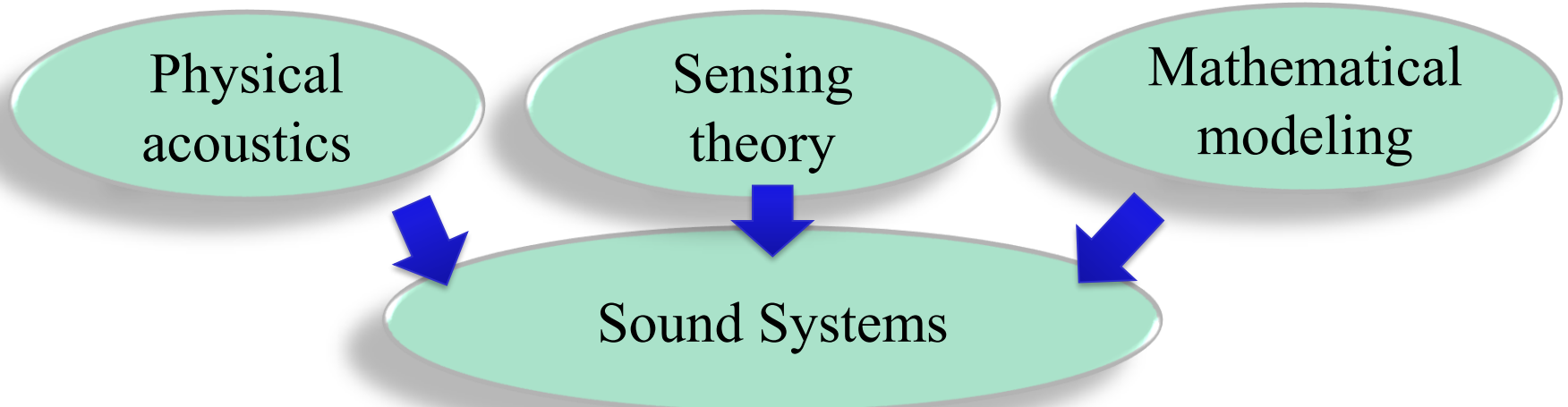


Overview of statistical speech & acoustic signal Processing

Why focus on sound media?

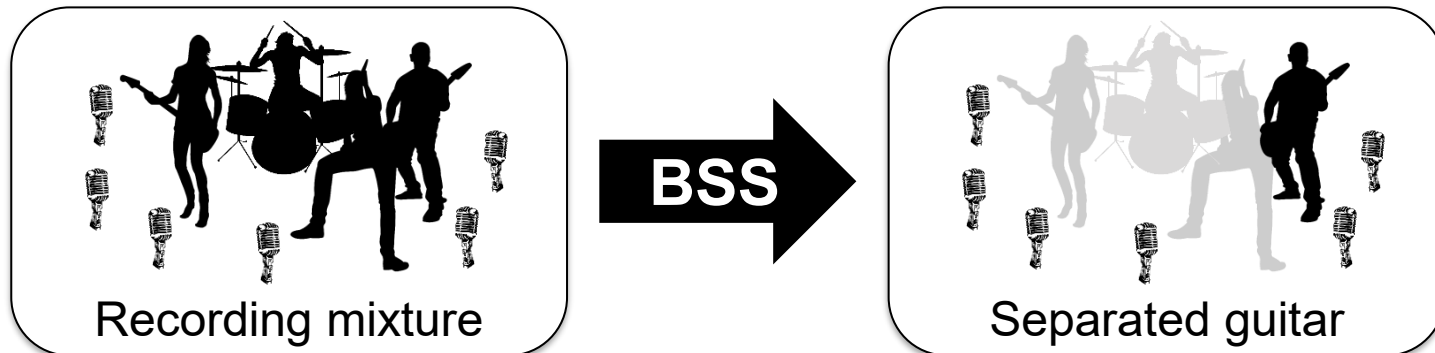
- What makes signal processing research related to audio media so appealing?
 - The infinite diversity of sounds in nature
 - The research approach has multiple faces (deterministic? statistical?).
 - The evaluation comes down to how well it's heard $\Rightarrow$ It also possesses artistic merit.

Main topics in lecture



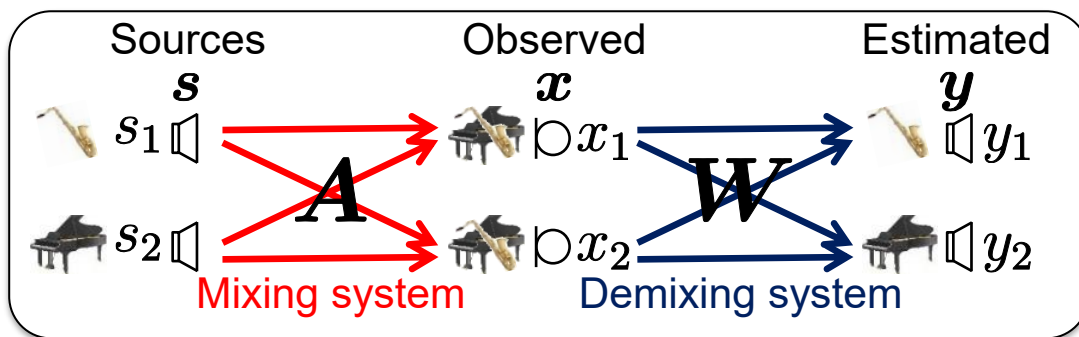
Ex: Blind source separation (BSS)

- Blind source separation (BSS) for audio signals

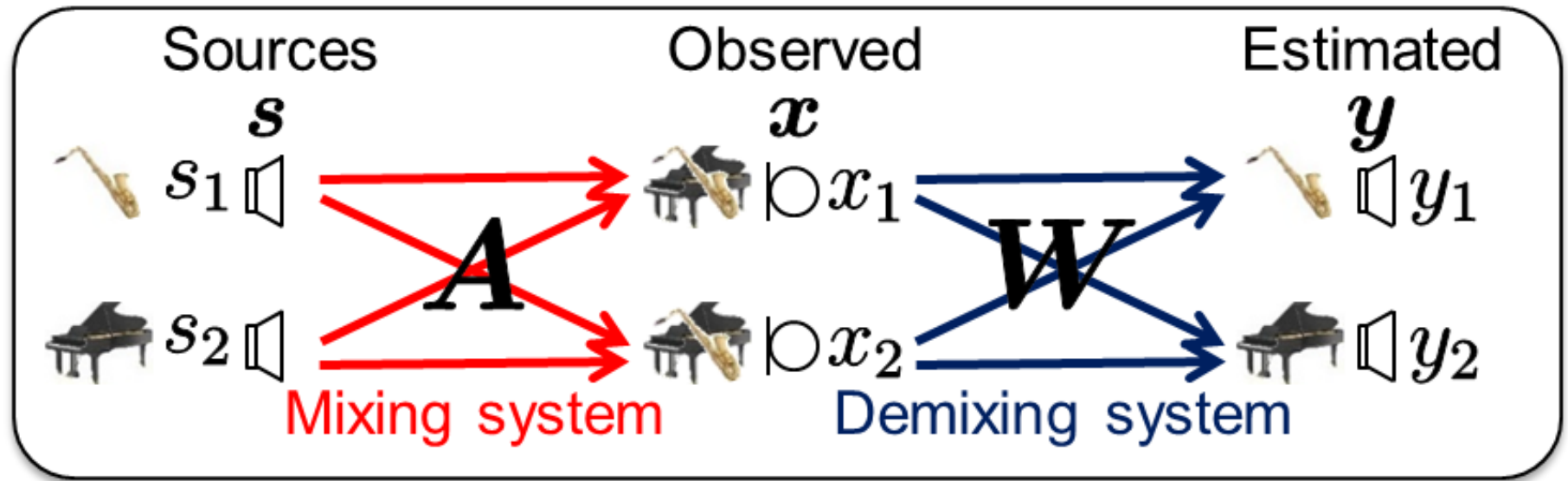


- separates original audio sources
 - does not require prior information of recording conditions
 - locations of mics and sources, room geometry, timbres, etc.
 - can be available for many audio applications
- Consider only “**determined**” situation

of mics
||
of sources



Why BSS is difficult and challenging?



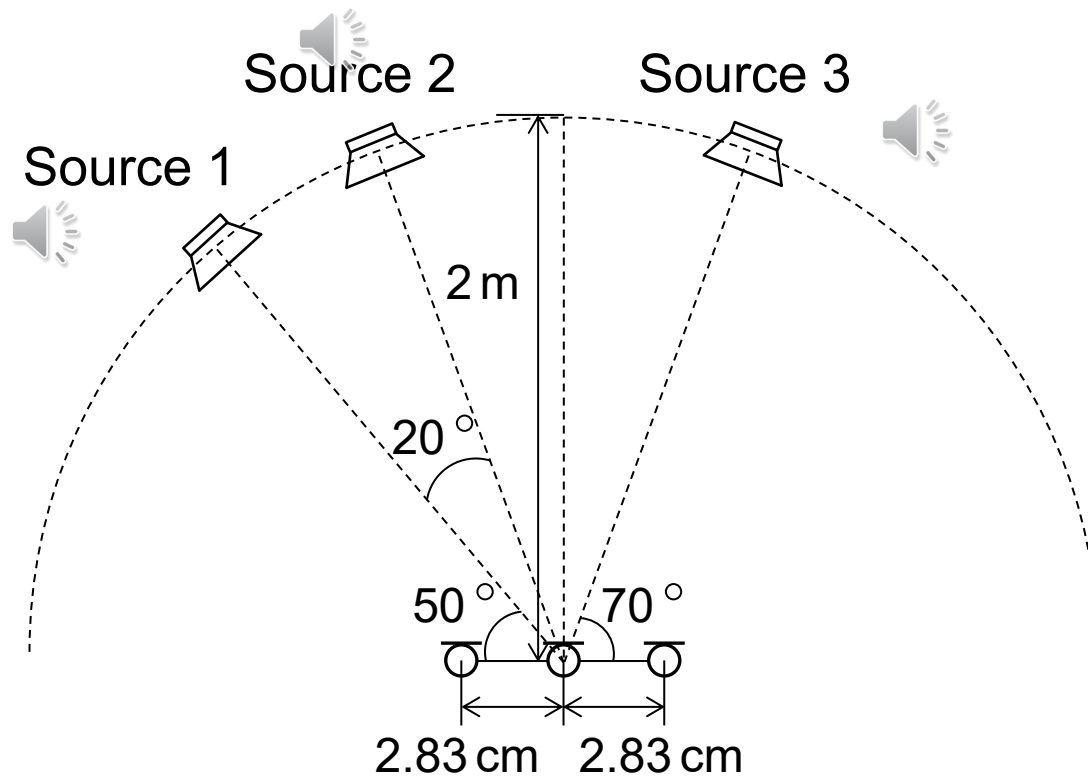
In BSS, we should estimate 2 groups of parameters.

- Spatial: Estimate W so that we maximize independence between outputs
- Source: Estimate p.d.f. $p(s)$ and spectral structure for each source

It is NOT convex problem and no closed-form solution!

Demonstration: music source separation

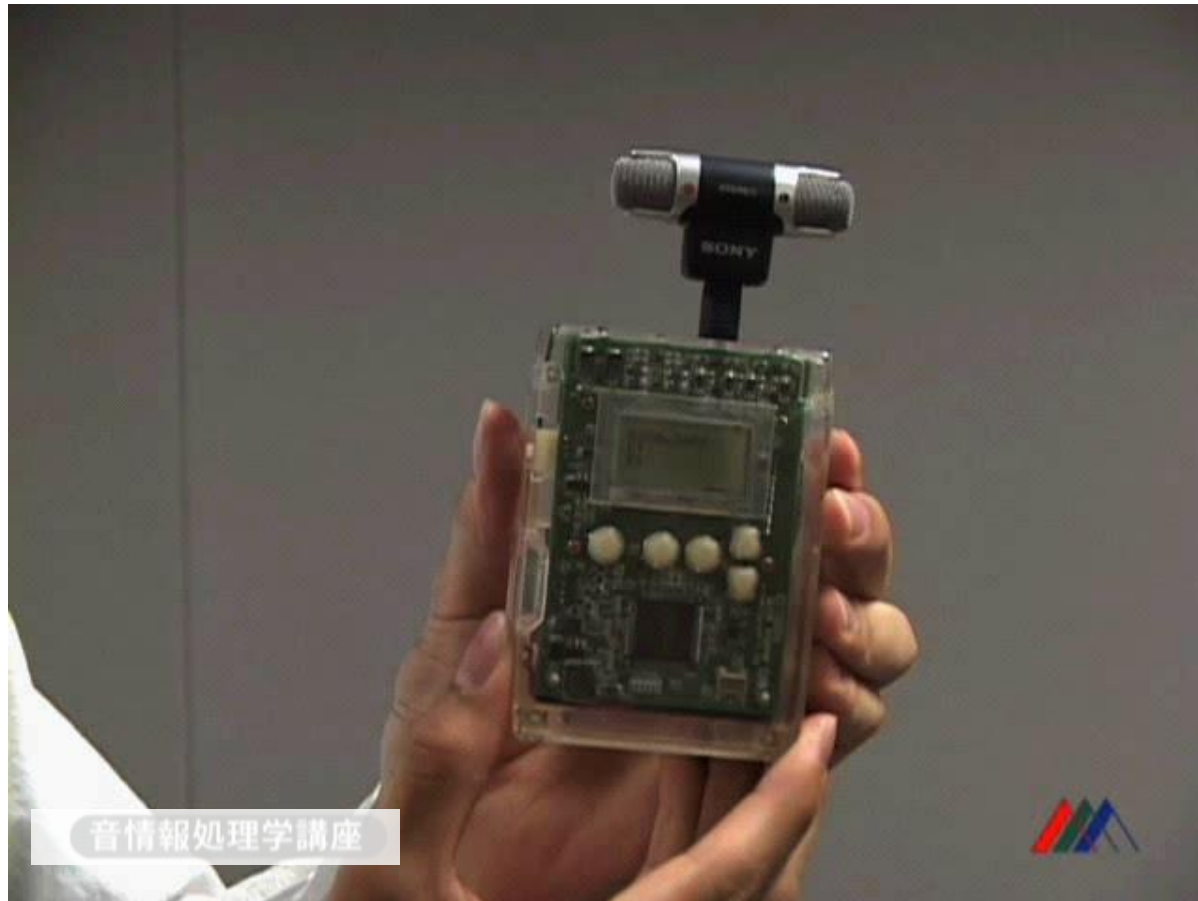
- Drum, Violin, Speech separation
(impulsive) (harmonics) (complex)



Source and mic positions are unknown!

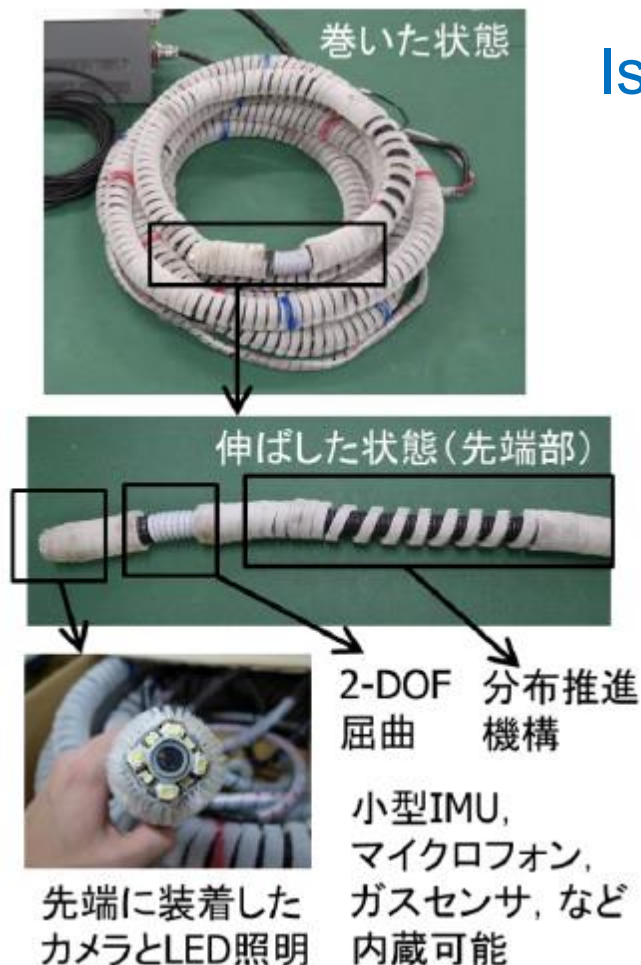
Realtime ICA Demo [Mori, Saruwatari, et al. 2009]

- In 2005, realtime ICA module with 4ch-mic and DSP (TI-C67) has been developed.
- In 2009, commercially available (Police Agency)



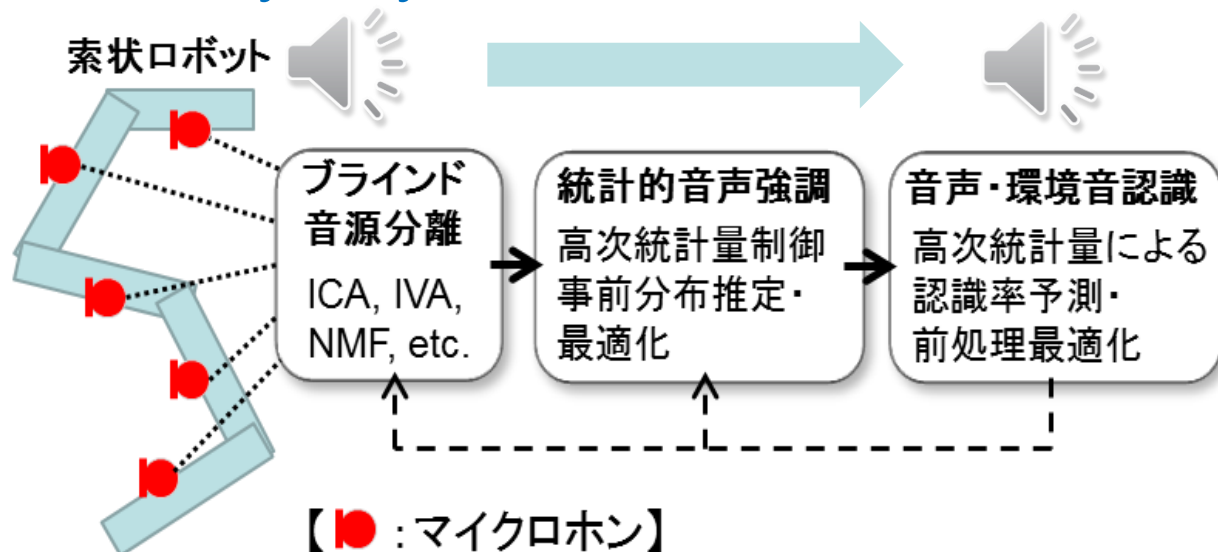
Rescue robot with BSS (Japan Cabinet Office Project)

- Snake-like rescue robot used in disaster area
- Microphone array detects victim's voice.



Is there anybody here?

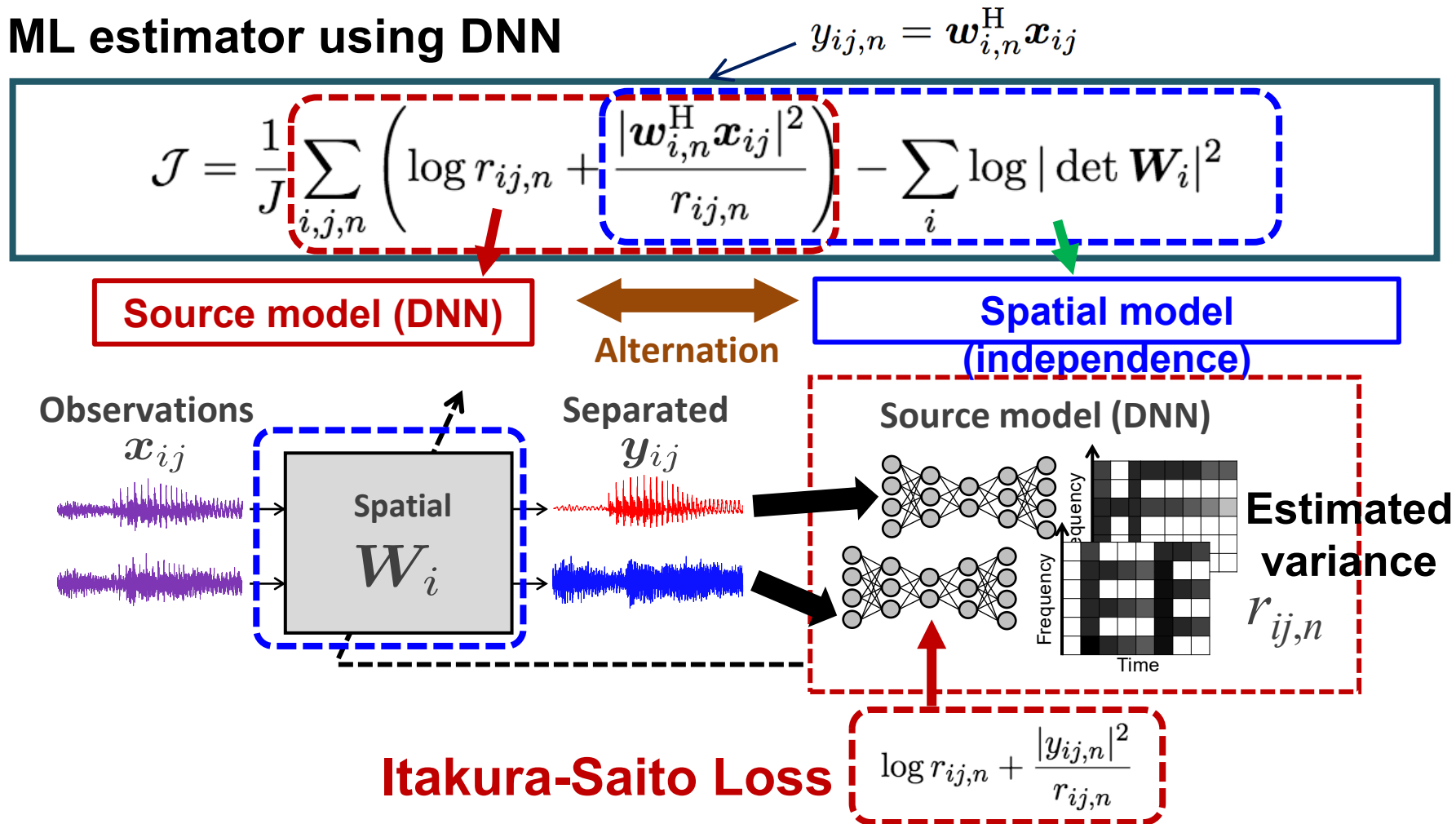
“Help me!”



Distributed microphone array (mic position is unknown) + ILRMA

Ex: Independent Deeply Learned Matrix Analysis (IDLMA)

■ ML estimator using DNN

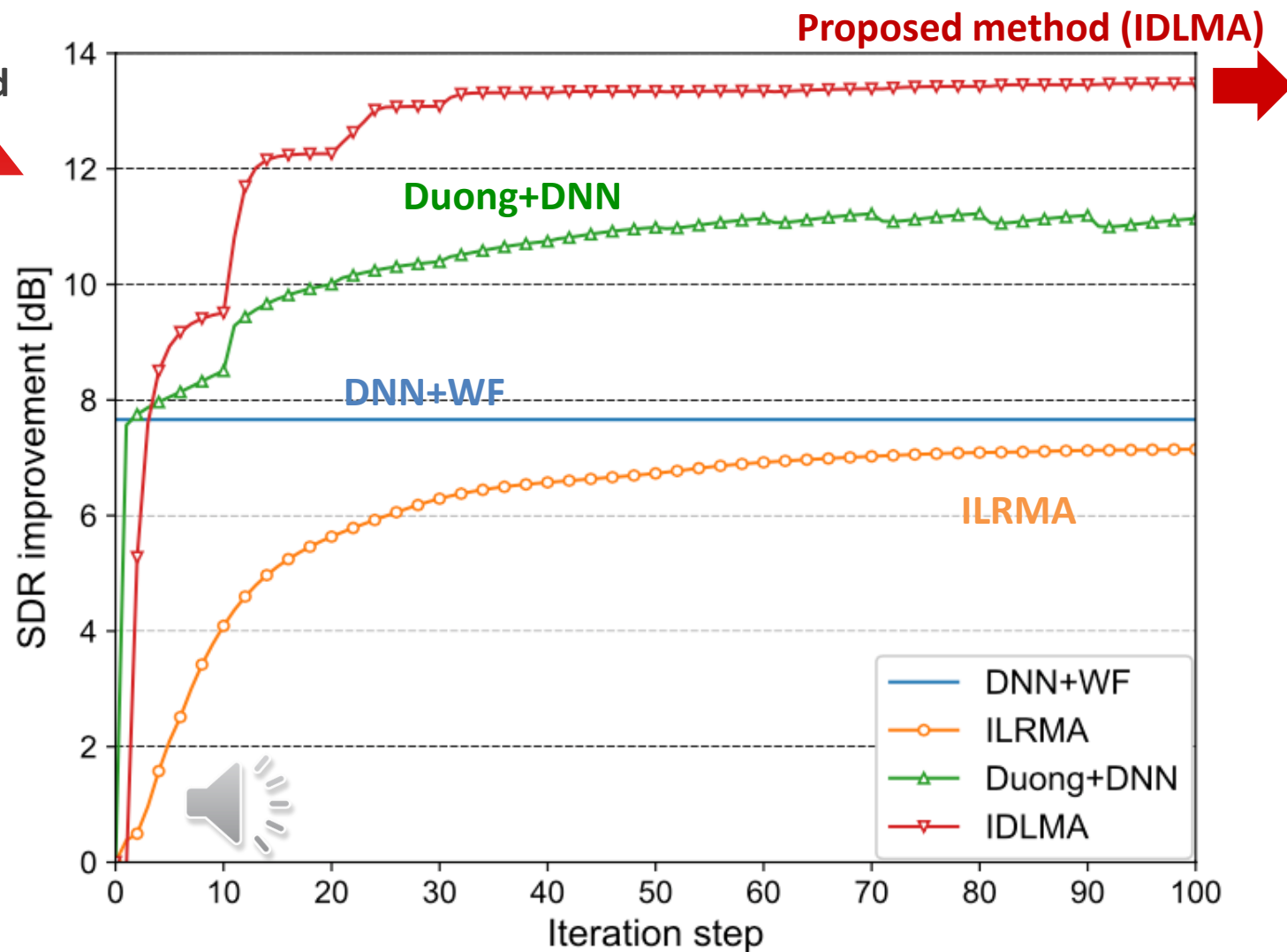


- **Spatial model** estimates $\mathbf{W}$ so that sources are mutually independent.
- **Source model** estimates variance $r_{ij,n}$ so that $\mathcal{J}$ is minimized via DNN.

Experimental results



Good



Vo.



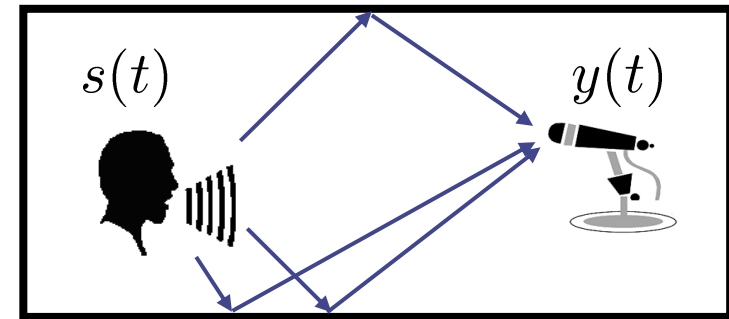
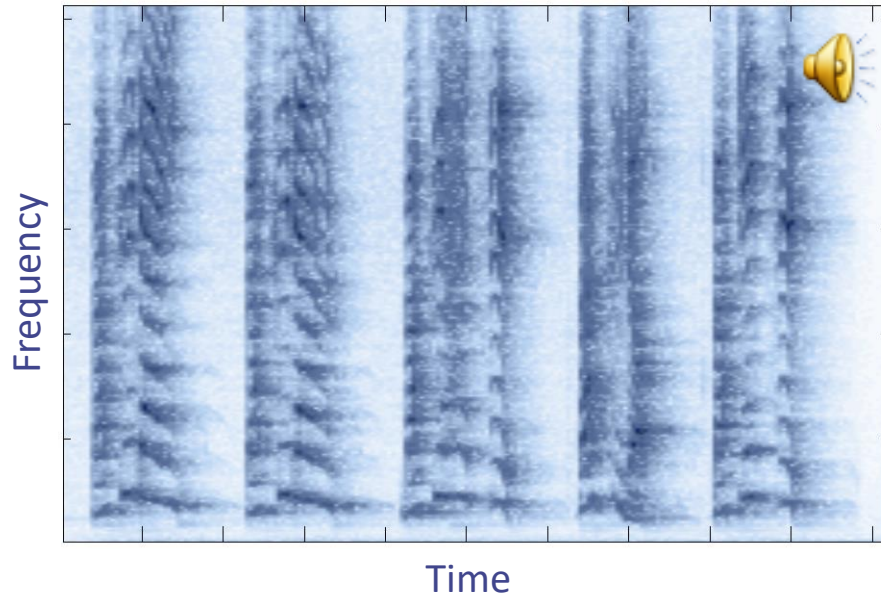
Ba.



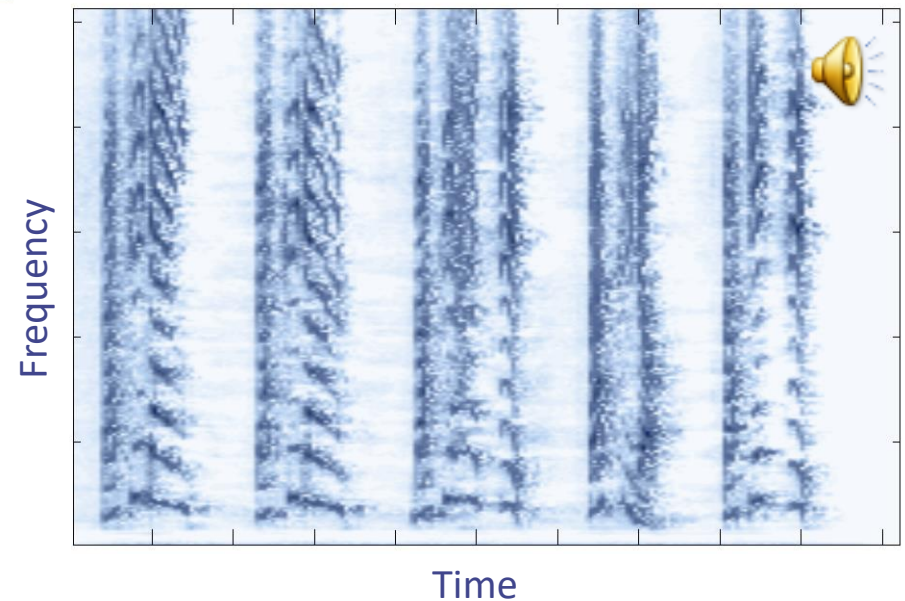
Ex: Reverberation suppression

[Kameoka, et al.]

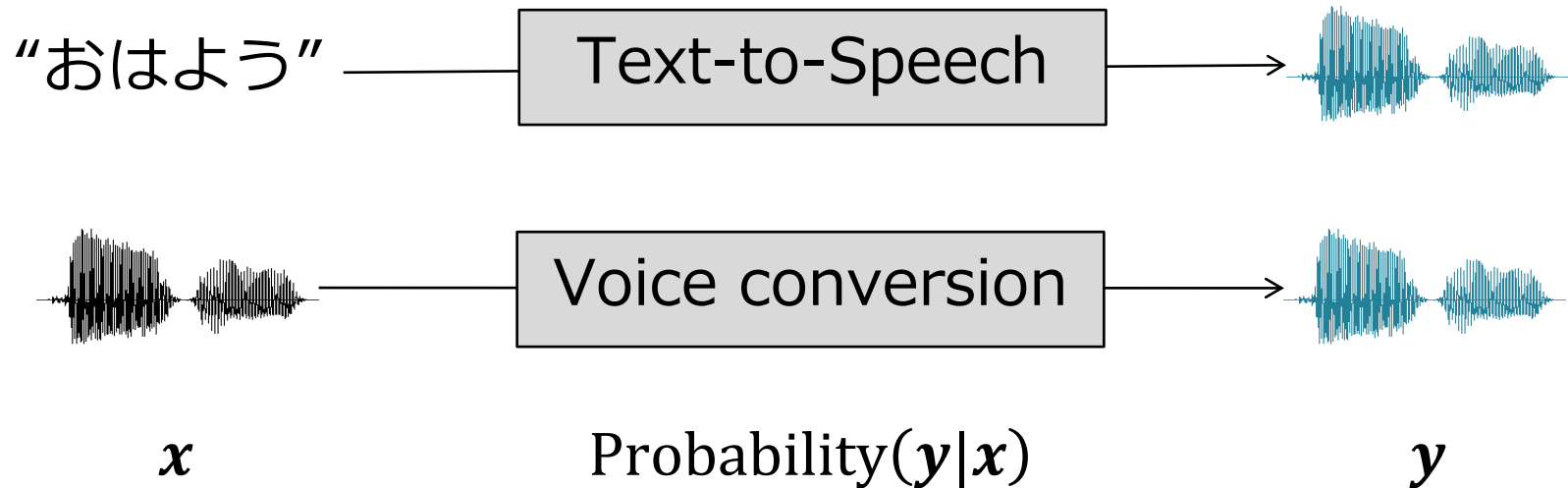
■ Input spectrogram with reverb



■ Output spectrogram with reverb suppression



Ex: Speech synthesis

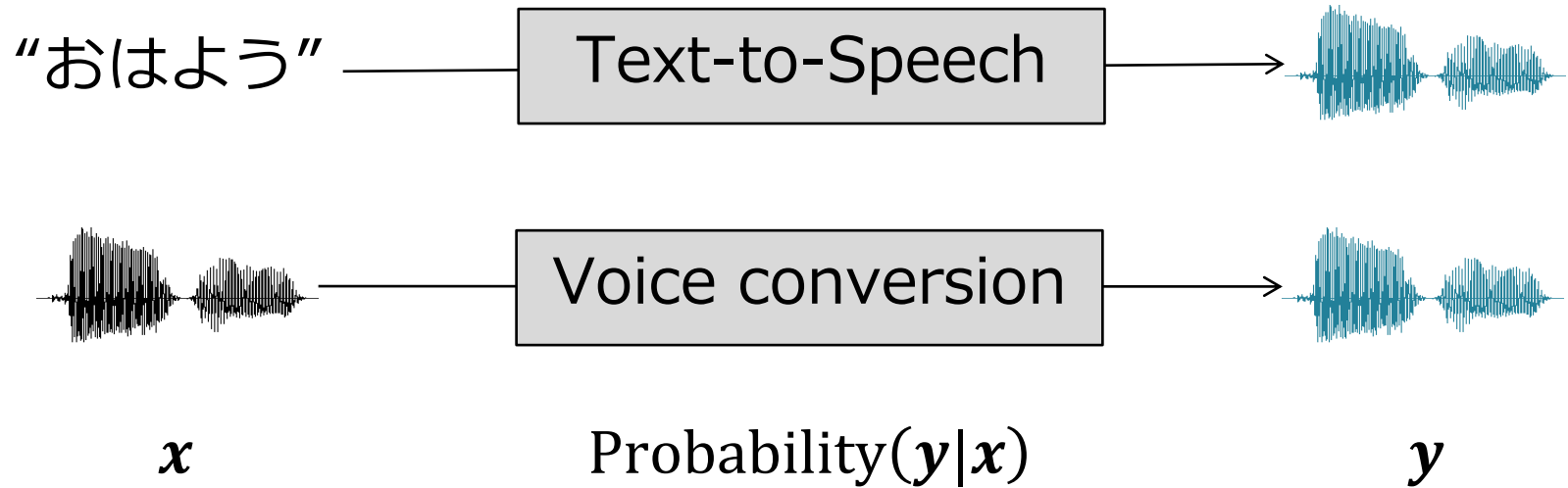


How to describe relation between Input x and output y ?

How to deal with variation in speech?

What contributes naturalness in human-like voice?

Ex: Speech synthesis



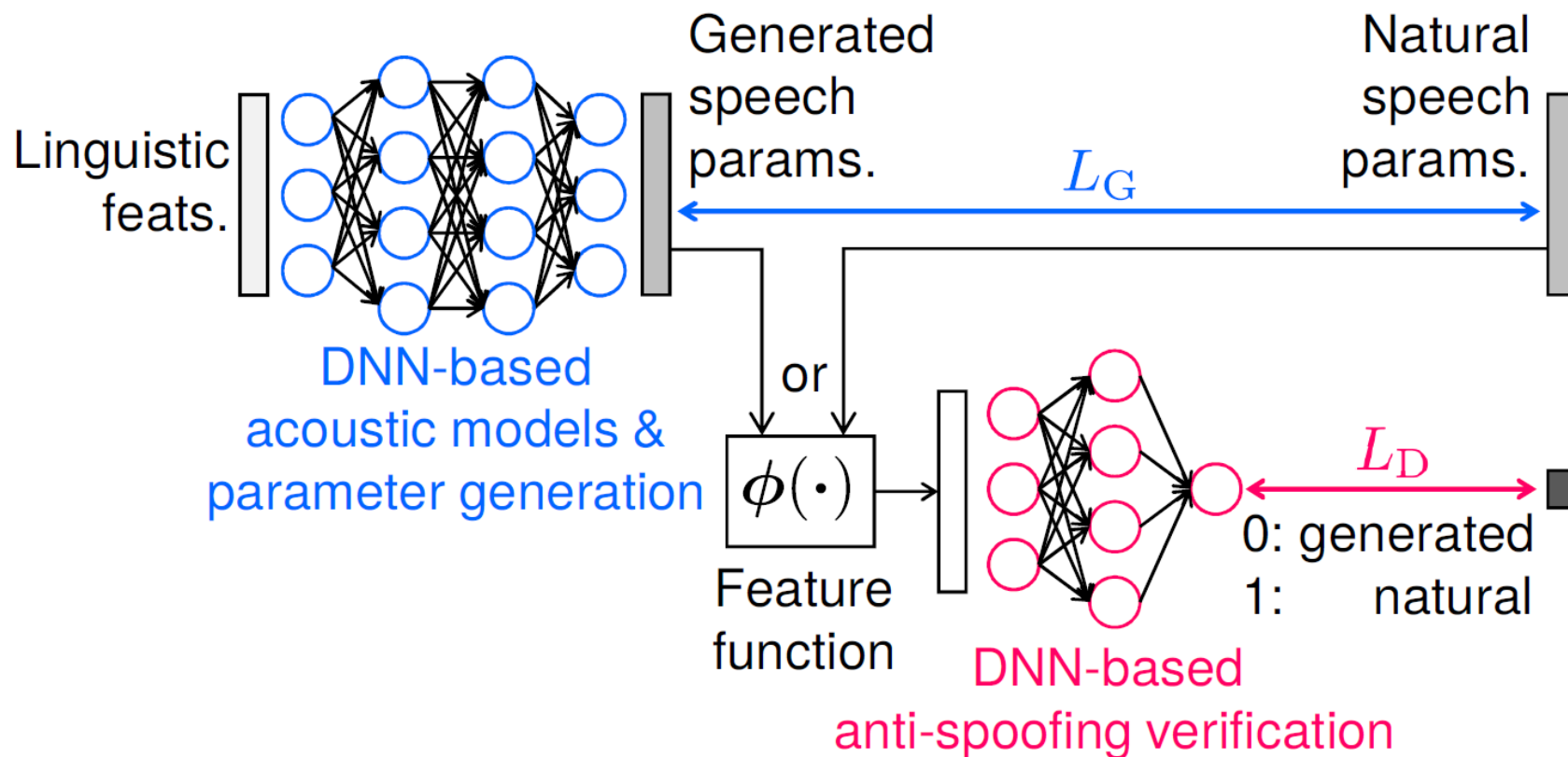
東京大学は世界一！
昼飯はとんこつラーメンに限る！





World-first GAN-based DNN TTS

[Saito, et al.; ICASSP2017, IEEE TASLP2018]



Become
human voice



Battle to avoid
being deceived





World-first GAN-based DNN TTS

[Saito, et al.; ICASSP2017, IEEE TASLP2018]



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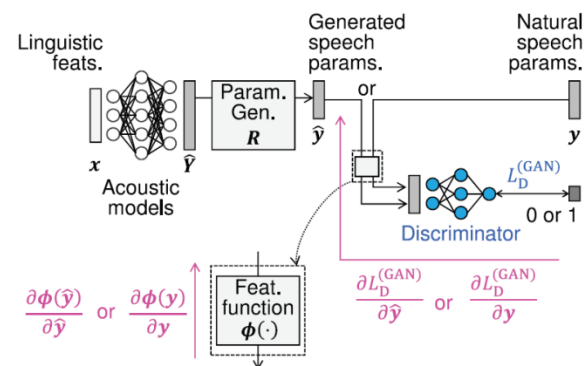
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Loss function and gradients for updating the discriminator. Param. Gen. indicates the speech parameter generation. For more, see "Statistical Parametric Speech Synthesis Incorporating Generative Adversarial Networks" by Y. Saito *et al.*, p. 84.

For the January 2018 issue, see p. 1 for Table of Contents

For the February 2018 issue, see p. 210 for Table of Contents



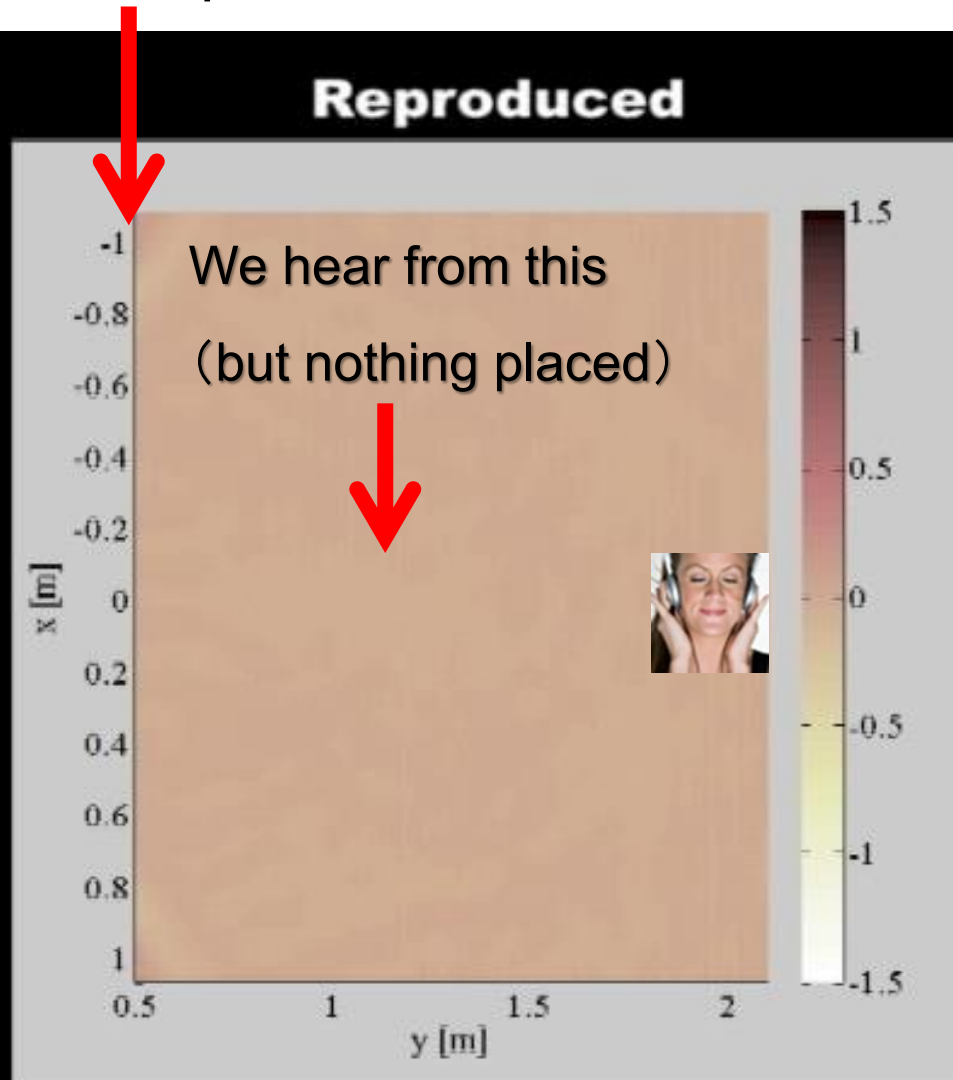
ロボット対話 デモンストレーション

Demonstration for
spoken-oriented hands-free
interactive robot.

Ex: Sound virtual reality

[Koyama, et al. 2014]

Real loudspeakers





Basis of statistical signal processing

Probability law

- Joint probability (or combination probability)

- $p(x, y)$: Probability that events x and y occur simultaneously.

- Conditional probability

- $p(y|x)$: Probability that y occurs under x has occurred.

$$p(x, y) = p(y|x)p(x) = p(x|y)p(y)$$

- Marginalization

- $p(y) = \int p(x, y)dx$

- Independence

- x and y are independent $\Leftrightarrow p(x, y) = p(x)p(y)$

Bayes' theorem

- $$\underline{p(x|y)} = \frac{p(y|x)p(x)}{p(y)} = \frac{\underline{p(y|x)}\underline{p(x)}}{\int \underline{p(y|x)}\underline{p(x)}dx}$$

Probability models (likelihood & prior probability)

$$\blacksquare \quad \underline{p(\theta|Y)} = \frac{\underline{p(Y|\theta)}\underline{p(\theta)}}{\int_{\theta} \underline{p(Y|\theta)}\underline{p(\theta)}}$$

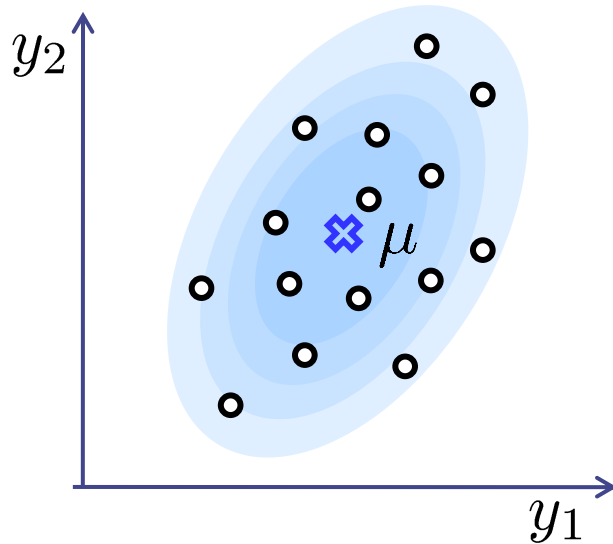
■ Let Y be the observed data and θ be the unknown parameter,

- $p(Y|\theta)$ is likelihood function,
- $p(\theta)$ is prior probability,
- $p(\theta|Y)$ is posterior probability.

Probability models

■ Probabilistic generation of data $y: p(\mathbf{y}|\boldsymbol{\theta})$

Ex 1) Normal distribution

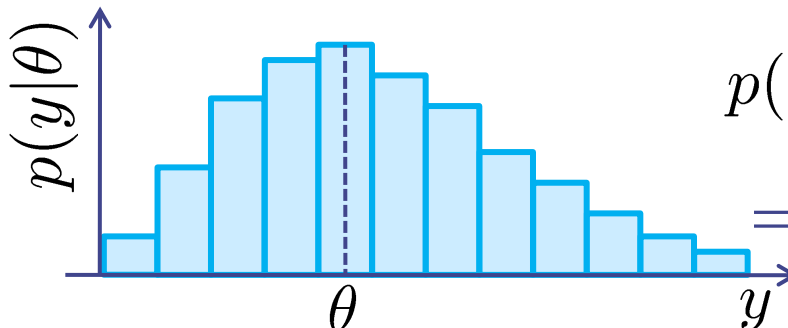


$\{\boldsymbol{\mu}, \boldsymbol{\Sigma}\}$ Mean, Variance

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^K |\boldsymbol{\Sigma}|}} e^{-\frac{1}{2}(\mathbf{y}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y}-\boldsymbol{\mu})}$$

$\Rightarrow$ we refer to as $\mathcal{N}(\mathbf{y}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

Ex 2) Poisson distribution



$$p(y|\theta) = \frac{\theta^y e^{-\theta}}{y!} \quad (y \in \mathbb{N})$$

$\Rightarrow$ we refer to as $\text{Pois}(y; \theta)$

Properties of normal distribution

- $x \sim \mathcal{N}(x; \mu_x, \Sigma_x)$, $y \sim \mathcal{N}(y; \mu_y, \Sigma_y)$ and x and y are independent
 - $z = x + y \Rightarrow z \sim \mathcal{N}(\mu_x + \mu_y, \Sigma_x + \Sigma_y)$
 - $z = Ax + b \Rightarrow z \sim \mathcal{N}(z; A\mu_x + b, A\Sigma_x A^\top)$
- $\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}\right)$
- $\mathbb{E}[x|y] = \mu_x + \Sigma_{xy}\Sigma_{yy}^+(y - \mu_y)$

ML, MAP, and MMSE estimators

■ Given data y , we can estimate parameter θ using

■ Maximum Likelihood (ML) estimator

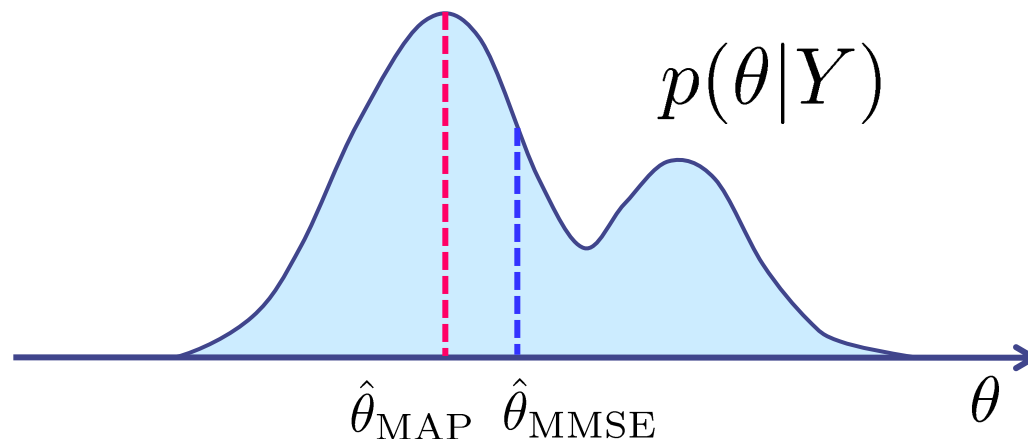
$$\hat{\theta}_{\text{ML}} = \underset{\theta}{\operatorname{argmax}} p(Y|\theta)$$

■ Maximum A Posteriori (MAP) estimator

$$\hat{\theta}_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}} p(\theta|y) = \underset{\theta}{\operatorname{argmax}} p(y|\theta)p(\theta)$$

■ Minimum Mean Squared Error (MMSE) estimator

$$\hat{\theta}_{\text{MMSE}} = \underset{\hat{\theta}}{\operatorname{argmin}} \mathbb{E}[(\hat{\theta} - \theta)^2|Y] = \mathbb{E}[\theta|Y]$$



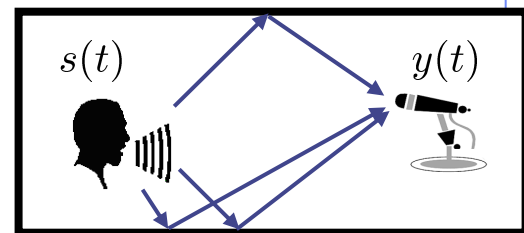
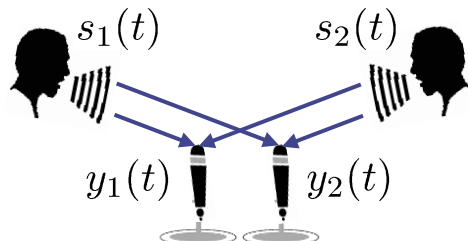
Many problems in acoustics are inverse problem

■ Acoustic signal processing

■ BSS

■ Dereverberation

...



■ Speech processing

■ Feature extraction

■ Speech recognition

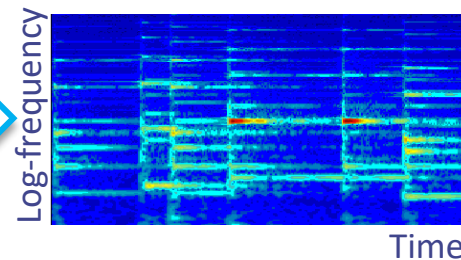
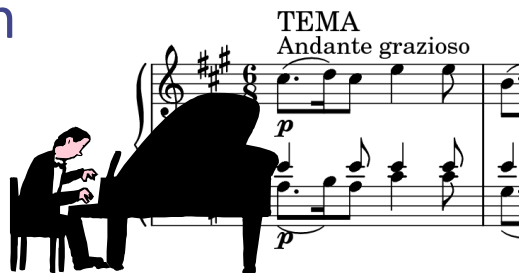
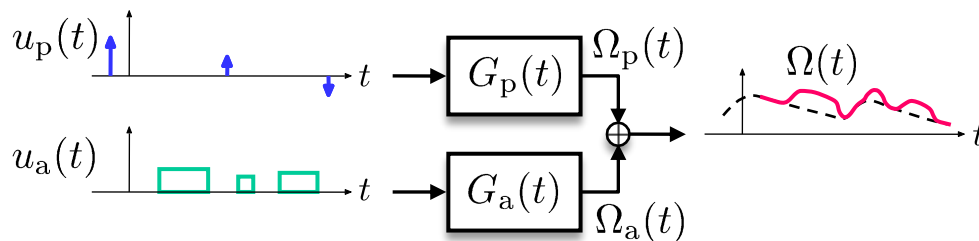
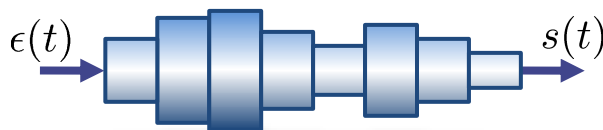
■ Intonation analysis

■ Music info. processing

■ Multitone analysis

■ Automatic transcription

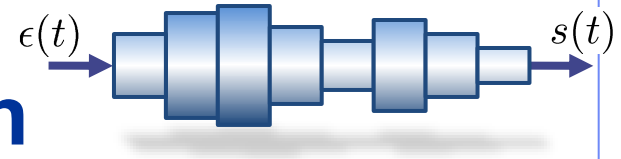
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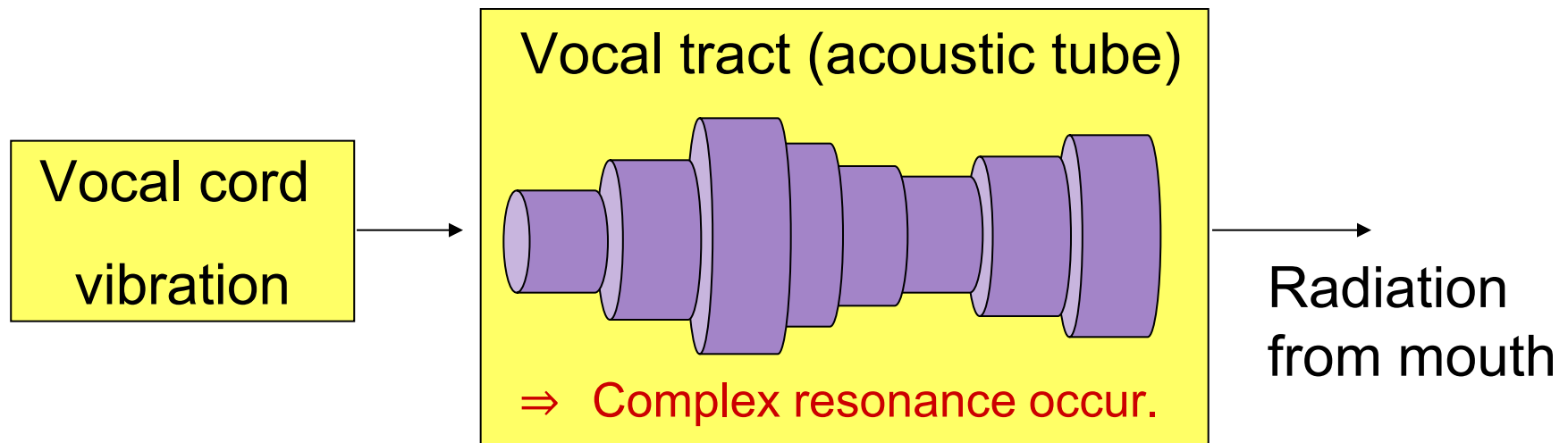
Linear Predictive Coding

Inverse problem in mobile phone



■ Generative model in human

- Vibrations in vocal cords are shaped into timbre in vocal tract.
- Vocal tract is a series of acoustic tubes with varying diameters → **autoregressive (AR) model**



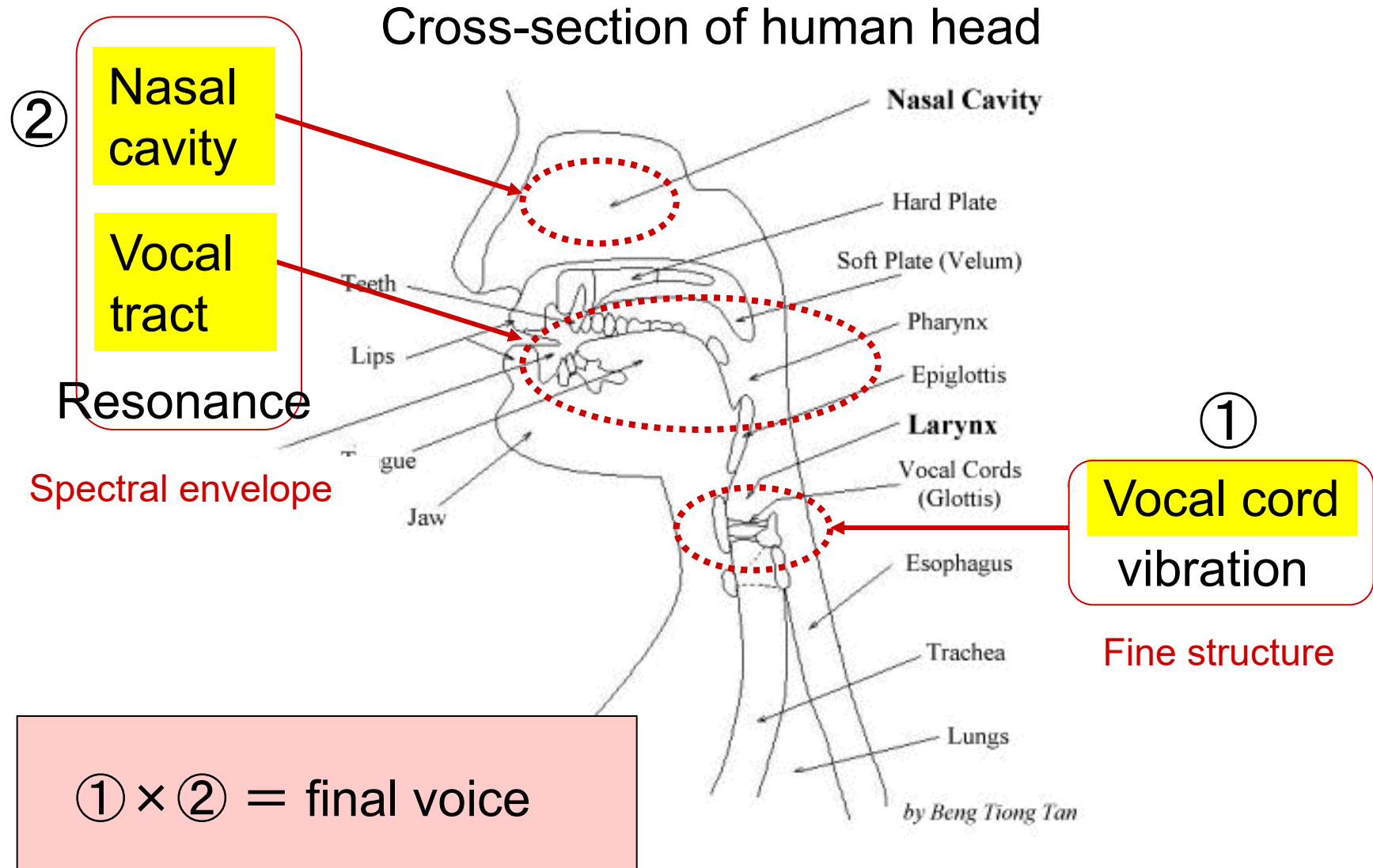
The generative AR model parameters are transmitted, resulting in significant information compression.

Spectral structure in speech

- Two elements of speech spectral structure
 - **Spectral envelope** that changes slowly with frequency
⇒ Resonance and anti-resonance characteristics of the vocal organs
(i.e., characteristic quantity that represents shape of human throat and mouth)
 - **Fine structure** that rapidly changes with periodic and non-periodic frequencies
⇒ Periodicity of sound sources
(i.e., characteristic that represents basic cycle of vocal cords and pitch of voice)

The spectrum of speech signal is represented by product of these two elements.

Spectral structure in speech



Linear prediction & ML estimation

- Explained from three perspectives
 - From perspective of minimizing “prediction error”
 - Least Squares Error Estimation
 - From perspective of linear system
 - AR system
 - Speech generation process model
 - ML estimation in time domain
 - Whitening
 - From perspective of spectral matching
 - ML estimation in frequency domain
 - Itakura-Saito Divergence

Minimize "prediction error"

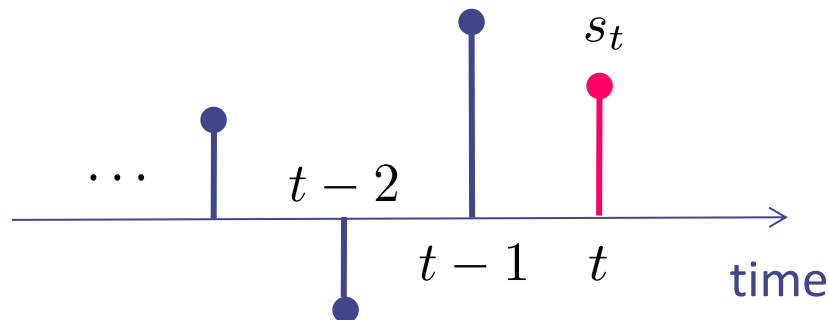
- Motivation: application to SPEECH CODING

- We want to represent speech with small number of parameters.

- Problem: Minimizing linear prediction error

- Sample s_t at time t is predicted by using liner combination of $s_{t-1}, s_{t-2}, \dots, s_{t-P}$
 - How should the coupling coefficients (also called *prediction coefficients*) be set to minimize the error in the "prediction"?

$$s_t \simeq \sum_{p=1}^P \underline{a_p} s_{t-p}$$



Formulation using least squares error estimation

■ At all t , we want to optimize $a_1, \dots, a_P$ so that $s_t \simeq \sum_{p=1}^P a_p s_{t-p}$

■ Cost function

$$\mathcal{J}(a_1, \dots, a_P) = \sum_t \left(s_t - \sum_{p=1}^P a_p s_{t-p} \right)^2 \rightarrow \text{minimize}$$

■ The optimal solution is given by $\frac{\partial \mathcal{J}}{\partial a_1} = \frac{\partial \mathcal{J}}{\partial a_2} = \dots = \frac{\partial \mathcal{J}}{\partial a_P} = 0$

$$\begin{aligned} \longrightarrow \frac{\partial \mathcal{J}}{\partial a_q} &= 2 \sum_t \left(s_t - \sum_{p=1}^P a_p s_{t-p} \right) (-s_{t-q}) \\ &= 2 \left(\sum_p a_p \sum_t s_{t-p} s_{t-q} - \sum_t s_t s_{t-q} \right) \\ &= 0 \quad (q = 1, \dots, P) \end{aligned}$$

Formulation using least squares error estimation

■ It reduces to a system of linear equations.

$$\left\{ \begin{array}{l} \sum_p a_p \sum_t s_{t-p} s_{t-1} = \sum_t s_t s_{t-1} \\ \sum_p a_p \sum_t s_{t-p} s_{t-2} = \sum_t s_t s_{t-2} \\ \vdots \\ \sum_p a_p \sum_t s_{t-p} s_{t-P} = \sum_t s_t s_{t-P} \end{array} \right. \longrightarrow c_{q,p} = \sum_t s_{t-p} s_{t-q}$$

The optimal prediction coefficient satisfies the following equation.

$$\begin{bmatrix} c_{1,1} & \cdots & c_{1,P} \\ \vdots & \ddots & \vdots \\ c_{P,1} & \cdots & c_{P,P} \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} c_{0,1} \\ \vdots \\ c_{0,P} \end{bmatrix}$$

This equation is called the *Yule-Walker equation*.

Levinson-Durbin (LD) algorithm

■ Yule-Walker equation

$$\begin{bmatrix} c_{1,1} & \cdots & c_{1,P} \\ \vdots & \ddots & \vdots \\ c_{P,1} & \cdots & c_{P,P} \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} c_{0,1} \\ \vdots \\ c_{0,P} \end{bmatrix}$$

$$c_{p,q} = v_{|p-q|}$$

$$\begin{bmatrix} v_0 & v_1 & \cdots & v_{P-1} \\ v_1 & v_0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & v_1 \\ v_{P-1} & \cdots & v_1 & v_0 \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} v_1 \\ \vdots \\ v_P \end{bmatrix}$$

Toeplitz matrix

How to solve?

- General case:
Gaussian elimination
- positive definite symmetric:
Cholesky decomposition
- Toeplitz matrix:
Levinson algorithm

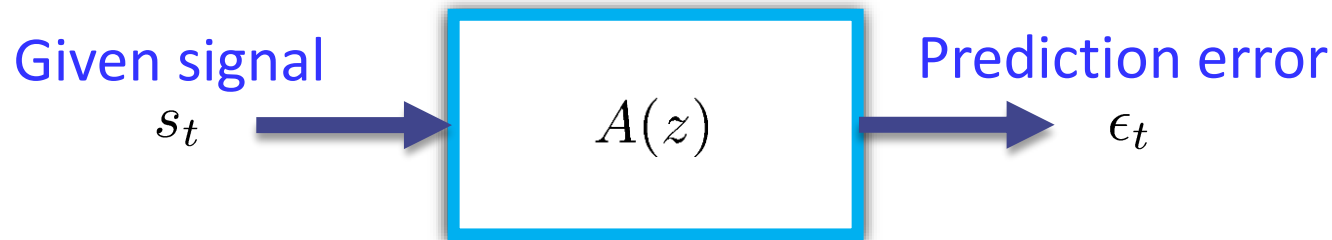
✂ Which one?

We can use fast algorithm
(LD algorithm)

Interpretation as a linear system

- Linear system that outputs prediction error from given signal

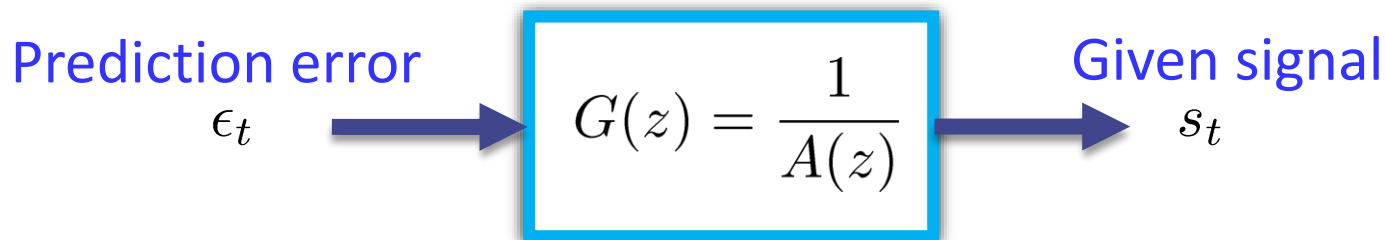
$$\epsilon_t = s_t - \sum_{p=1}^P a_p s_{t-p}$$



$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - \dots - a_P z^{-P}$$

Moving average system (all-zero model)

- Linear system that outputs given signal from prediction error?

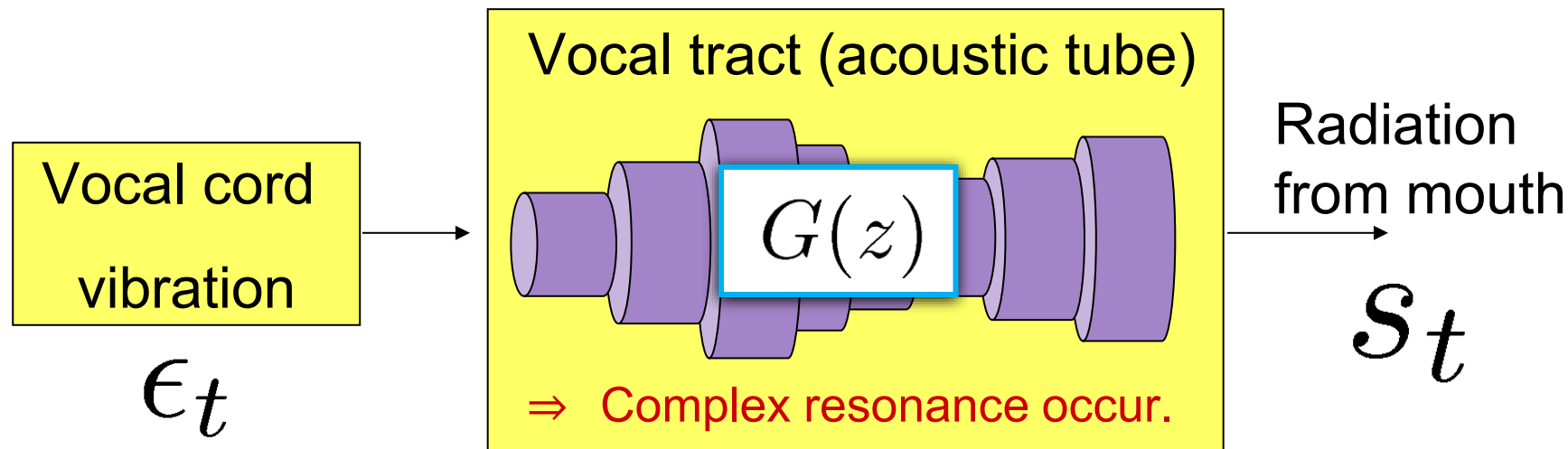
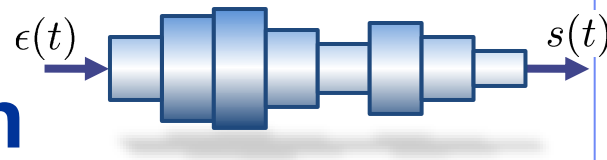


AR system (all-pole model)

System for speech inverse problem

■ Generative model in human

- Vibrations in vocal cords are shaped into timbre in vocal tract.
- Vocal tract is a series of acoustic tubes with varying diameters → AR model $G(z)$



Statistical modeling of speech production process

Assumptions regarding vocal cord vibration

- Gaussian $\epsilon = (\epsilon_1, \dots, \epsilon_T)^T \sim \mathcal{N}(\mu, \Sigma)$
 - Stationary $\mu = \mathbf{0}, \quad \Sigma : \text{Toeplitz matrix}$
 - Whiteness $\Sigma = \sigma^2 \mathbf{I}$
- } $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$
($t = 1, \dots, T$)

Assumptions regarding vocal tract characteristics

- AR system (all-pole model)

$$\underbrace{\begin{bmatrix} 1 & & & & 0 \\ -a_1 & 1 & & & \\ \vdots & \ddots & \ddots & & \\ -a_P & & \ddots & \ddots & \\ 0 & \ddots & & \ddots & \ddots \\ 0 & & -a_P & \cdots & -a_1 & 1 \end{bmatrix}}_{\Psi} \underbrace{\begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_T \end{bmatrix}}_{\mathbf{s}} = \underbrace{\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_T \end{bmatrix}}_{\epsilon}$$

Maximum likelihood estimation

■ Summary of assumptions

- $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$

- $\Psi \mathbf{s} = \epsilon$

Unknown parameters $\{a_p\}_{1 \leq p \leq P}, \sigma^2$

Observed signal $\{s_t\}_{1 \leq t \leq T}$

■ Probability density function (PDF) of observation (likelihood)

- $\mathbf{s} = \Psi^{-1} \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \Psi^{-1} (\Psi^{-1})^T)$

■ Log-likelihood is given by

$$-\frac{T}{2} \log(2\pi) - \frac{1}{2} \log |\sigma^2 \Psi^{-1} (\Psi^{-1})^T| - \frac{1}{2\sigma^2} \underbrace{\mathbf{s}^T \Psi^T \Psi \mathbf{s}}_{\rightarrow \|\epsilon\|_2^2} =$$

- Logdet term:
$$\begin{aligned} & \frac{\log |\sigma^2 \Psi^{-1} (\Psi^{-1})^T|}{=} \log |\sigma^2 \mathbf{I}| - 2 \log |\Psi| \\ & = T \log \sigma^2 \quad \because |\Psi| = 1 \end{aligned}$$

$$\sum_t \left(s_t - \sum_{p=1}^P a_p s_{t-p} \right)^2$$

Whitening effect: frequency-domain representation

- Since the above statistical model assumed whiteness for ε , the previous ML estimation was attempting to determine $\{a_p\}_{1 \leq p \leq P}$ such that ε would be as white as possible.
- To make this easier to visualize, let's formulate the above model in the frequency domain.

Formulation in frequency domain

■ In time domain, ...

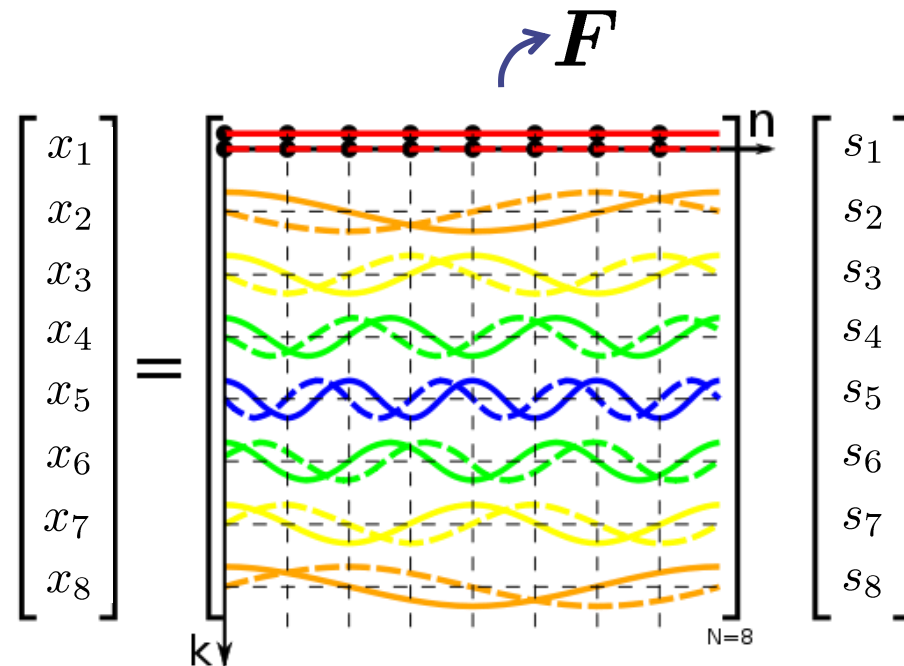
$$\left. \begin{array}{l} \blacksquare \Psi^{-1} \epsilon = s \\ \blacksquare \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \end{array} \right\} \longrightarrow s \sim \mathcal{N}(\mathbf{0}, \sigma^2 \Psi^{-1} (\Psi^{-1})^T)$$

■ In frequency domain (Fourier transform domain), ...

■ $x = F s$ (F is Fourier transform mat.): what is PDF of x ?

$$x \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2 \underline{F \Psi^{-1} (\Psi^{-1})^T F^H})$$

$$\begin{aligned} & F \Psi^{-1} (\Psi^{-1})^T F^H \\ &= (F \underline{\Psi^T \Psi} F^H)^{-1} \end{aligned}$$



Property of $\Psi^T \Psi$

■ We assume the following cyclic matrix type on Ψ

$$\Psi = \begin{bmatrix} 1 & & & -a_P & \cdots & -a_1 \\ -a_1 & 1 & & & \ddots & \vdots \\ \vdots & \ddots & \ddots & & & -a_P \\ -a_P & & \ddots & \ddots & & \\ & \ddots & & \ddots & \ddots & \\ 0 & & -a_P & \cdots & -a_1 & 1 \end{bmatrix}$$

■ Therefore, $\Psi^T \Psi$ is also cyclic matrix.

→ $\Psi^T \Psi$ can be diagonalized by F

$F \Psi^T \Psi F^H$: diagonal matrix

$$\sigma^2(F \Psi^T \Psi F^H)^{-1} = \text{diag}(\lambda_1, \dots, \lambda_T)$$

$$\lambda_t = \frac{\sigma^2}{|A(e^{2\pi j(t-1)/T})|^2}$$

$$A(z) = 1 - a_1 z^{-1} - \cdots - a_P z^{-P}$$

PDF of frequency components

- To summarize the above,

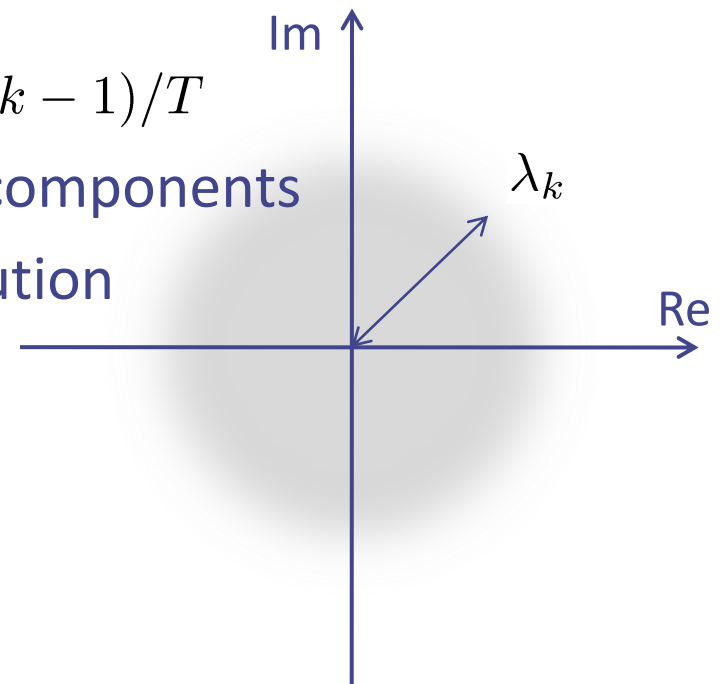
$$\mathbf{x} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \text{diag}(\lambda_1, \dots, \lambda_T))$$

$$\lambda_t = \frac{\sigma^2}{|A(e^{2\pi j(t-1)/T})|^2}$$

$$A(z) = 1 - a_1 z^{-1} - \dots - a_P z^{-P}$$

- x_k is component at frequency $2\pi(k-1)/T$
 - Independent of other frequency components
 - It follows complex normal distribution with variance λ_k

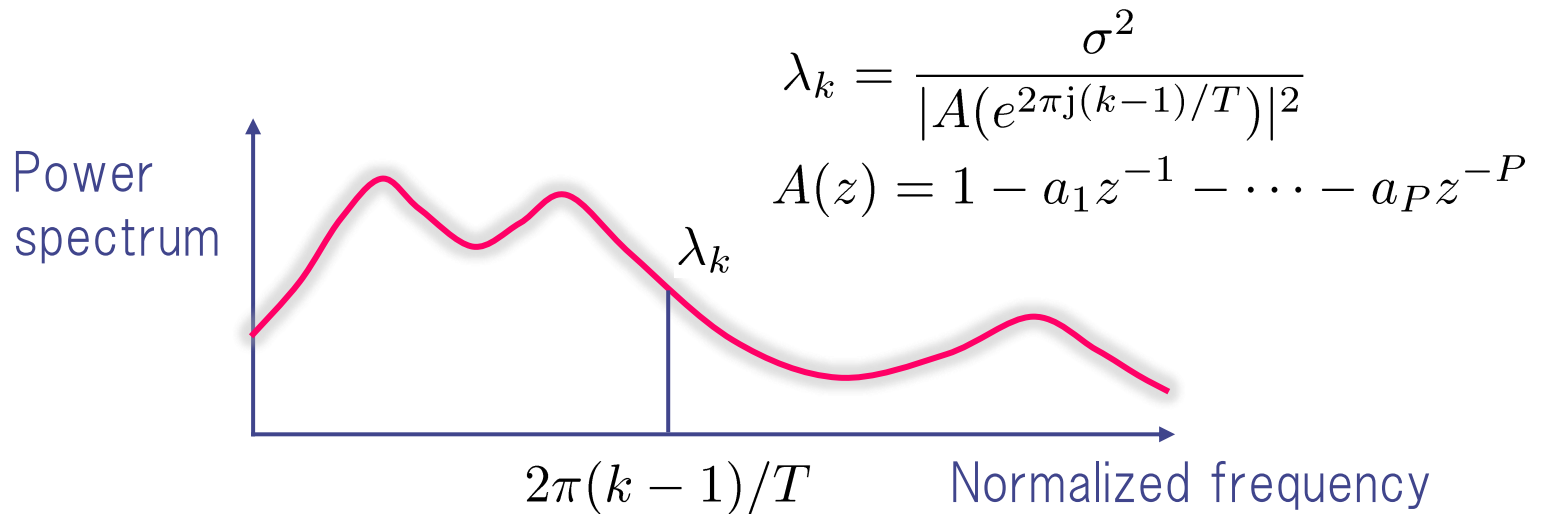
$$x_k \sim \mathcal{N}_{\mathbb{C}}(0, \lambda_k)$$



Viewpoint as "spectral matching"

- Log-likelihood given frequency components $\{x_k\}_{1 \leq k \leq T}$

$$\mathcal{L}(\lambda_1, \dots, \lambda_T) = - \sum_k \left(\log \pi \lambda_k + \frac{|x_k|^2}{\lambda_k} \right)$$



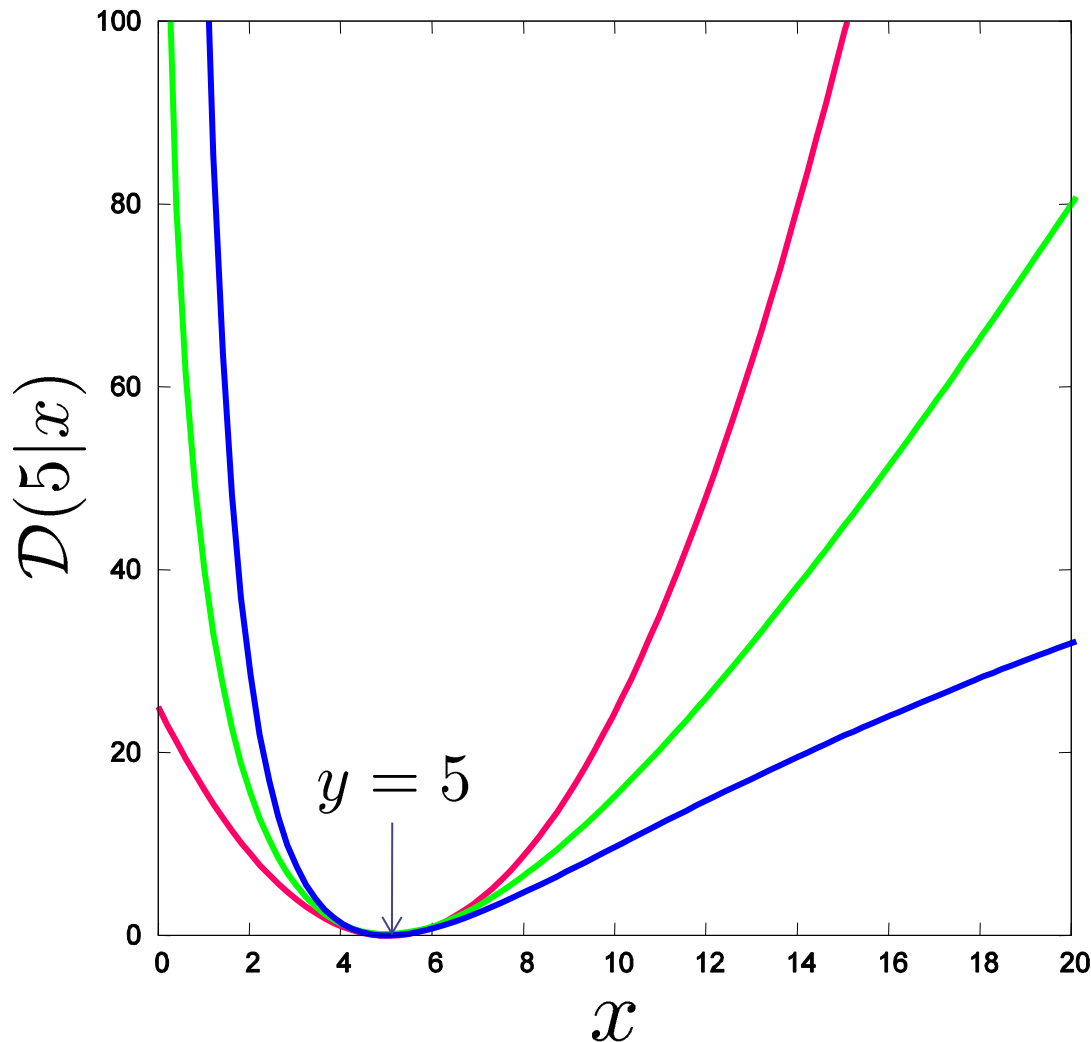
- The above log-likelihood is equal to the following, excluding the constant term:

$$\mathcal{L}(\lambda_1, \dots, \lambda_T) \stackrel{c}{=} - \sum_k \left(\frac{|x_k|^2}{\lambda_k} - \log \frac{|x_k|^2}{\lambda_k} - 1 \right)$$

Itakura-Saito divergence

Itakura-Saito divergence

■ Comparison with other distance scales



— Squared error

$$\mathcal{D}(x|y) = (y - x)^2$$

— I (KL) divergence

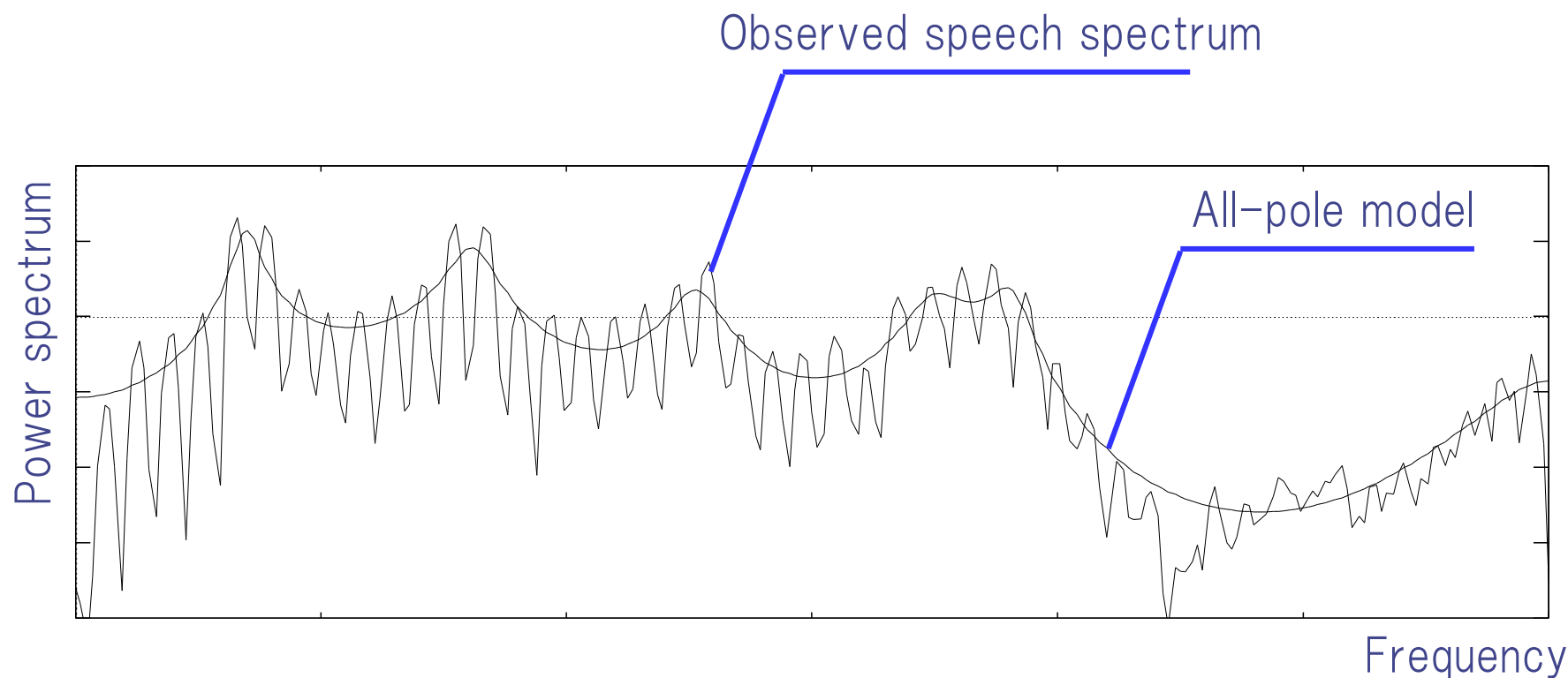
$$\mathcal{D}(x|y) = y \log \frac{y}{x} - (y - x)$$

— Itakura-Saito div.

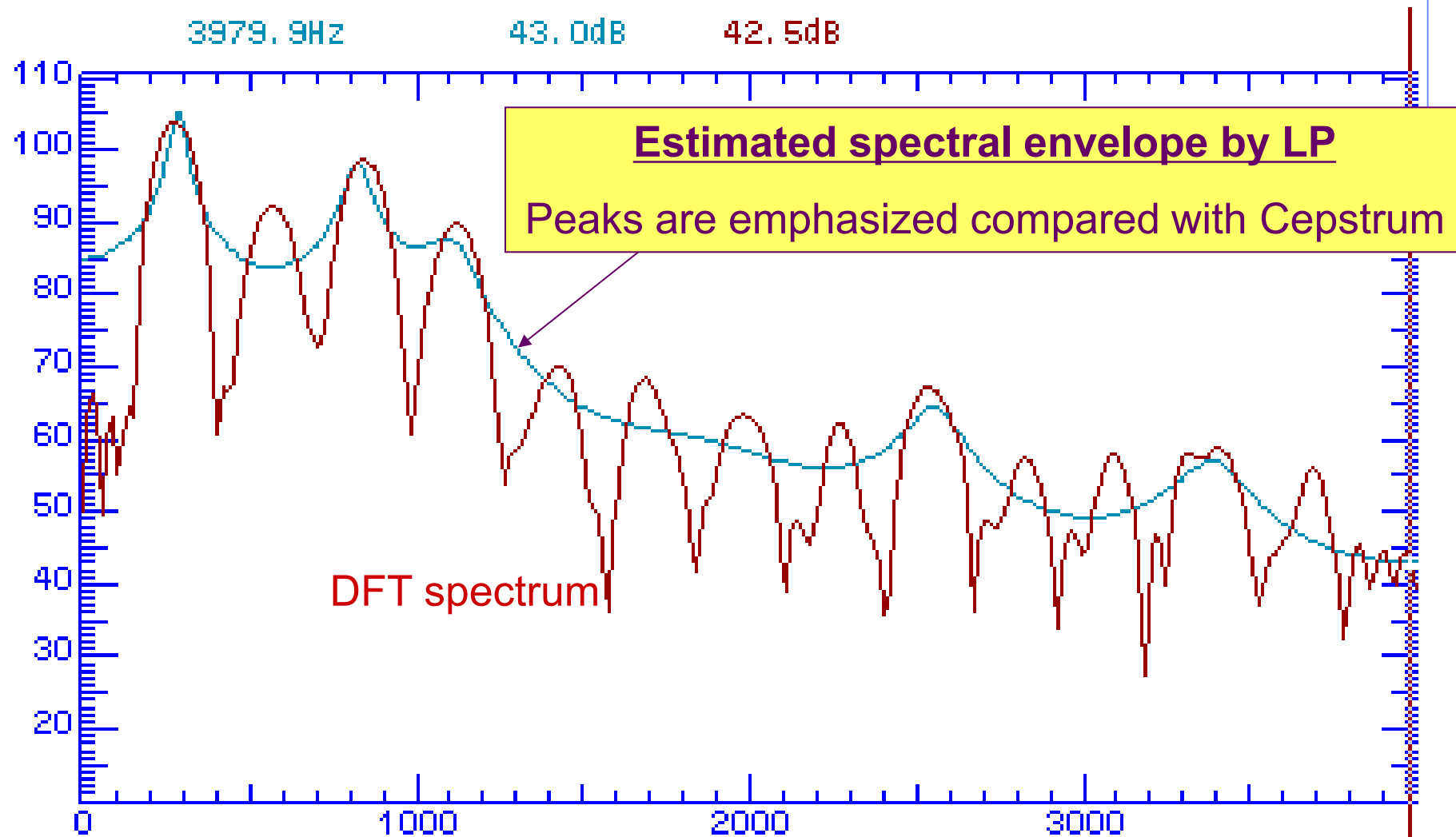
$$\mathcal{D}(x|y) = \frac{y}{x} - \log \frac{y}{x} - 1$$

Linear prediction as spectral envelope estimation

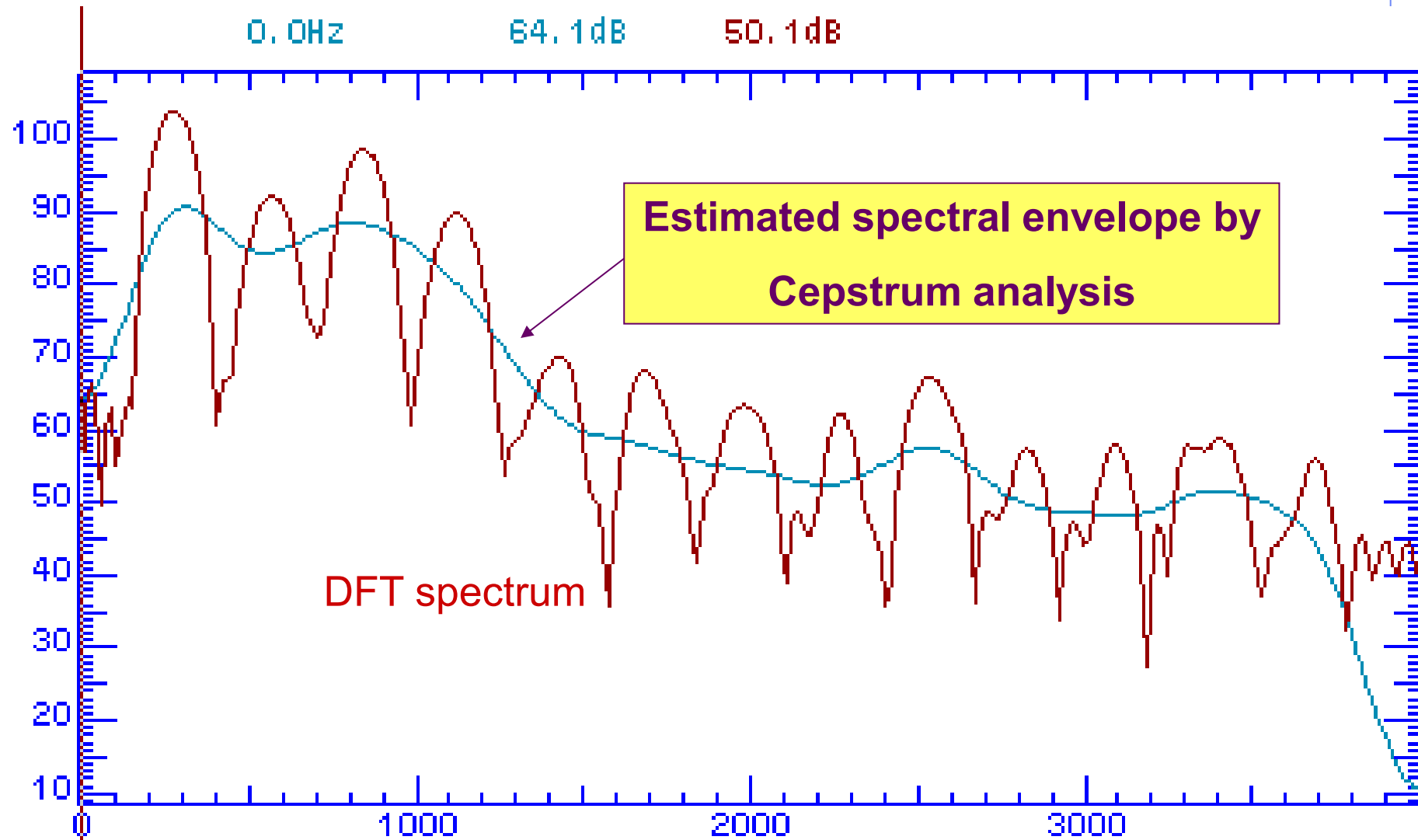
- Linear prediction is Itakura-Saito divergence minimization between observed power spectrum and all-pole-modeled spectrum.



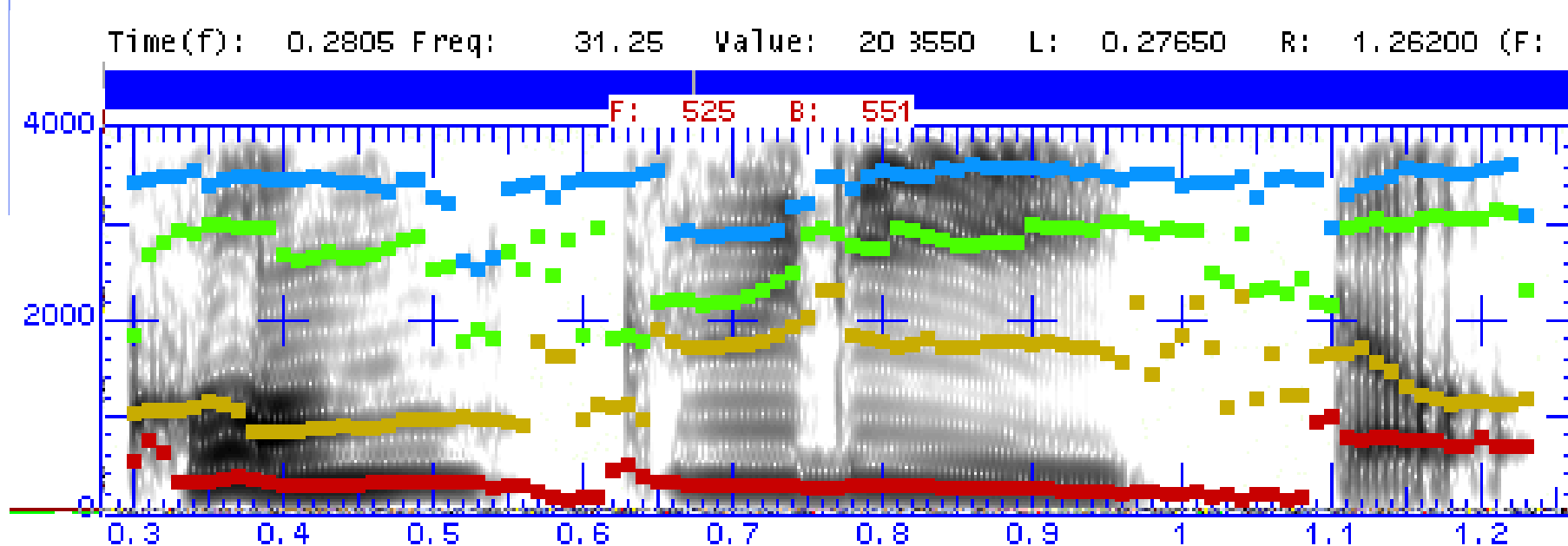
Advantages of spectral envelopes by linear prediction?



(Ref.) estimated spectral envelope by Cepstrum



Formant extraction by linear prediction



Linear prediction method & contribution of Japanese researcher

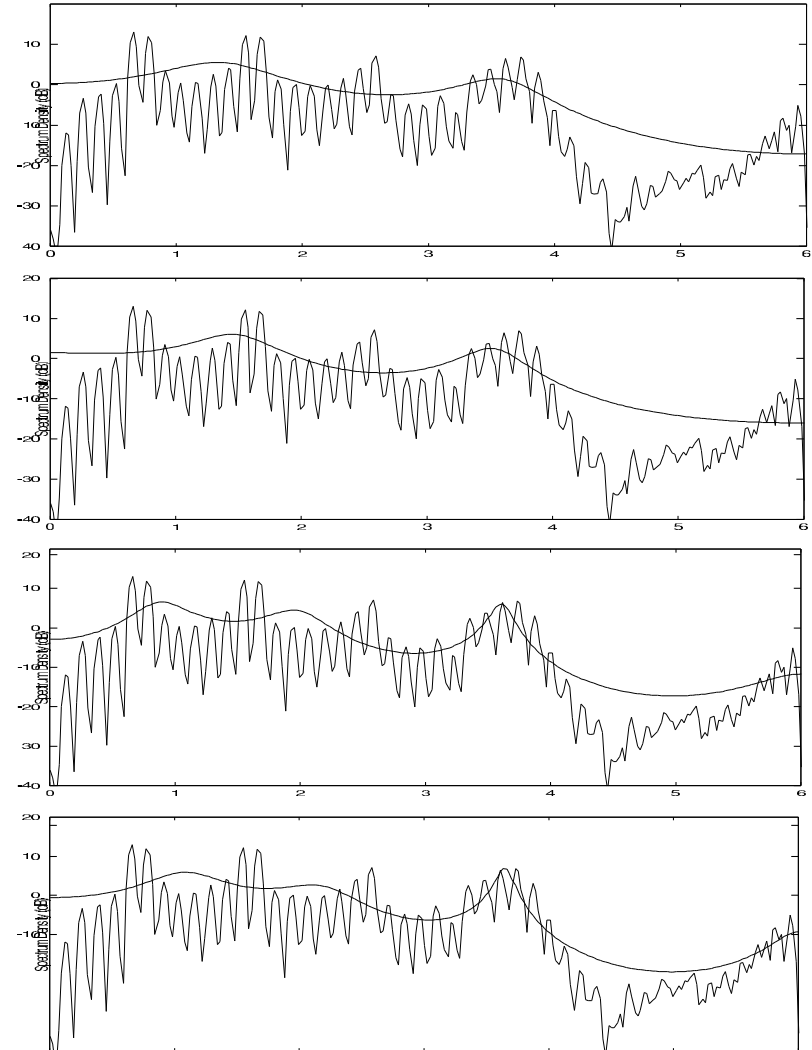
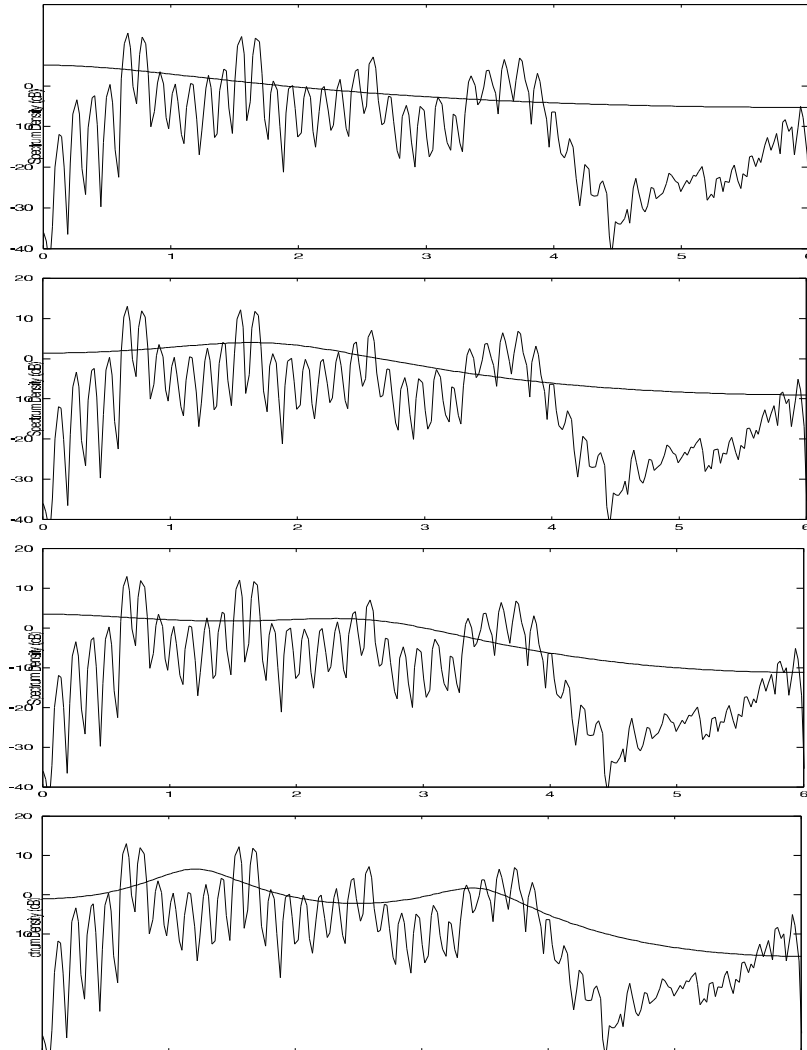
■日経産業新聞 1999年4月20日掲載

「音声認識の研究に金を出すことは価値ある投資だろうか。...音声認識の研究とはまさに「錬金術」に等しいー。」これは、1969年に米国音響学会誌に掲載された寄書の一部である。著者はジョン・ピアス、当時、ベル電話研究所情報通信部門の責任者であった。これを機に、**ピアス傘下にあった音声研究部門では、音声認識の研究が全面的に中止された。**

トップの確信に満ちた判断で中止された研究をボトムアップで再開するのは容易なことではない。**この再開の主役として登場したのは1人の日本人であった。線形予測理論で世界的脚光を浴びていた現NTTの板倉文忠(名古屋大名誉教授)を客員研究員として招いた。**これがベル研での音声認識研究の再始動をうながしたのである。当時、ベル研の研究室長であり、板倉を招いたジェームス・フラナガン(現米国ラトガース大学副学長)は振り返る。**「とても、正面切って音声認識の研究を行える状況ではなかった。部外者である客員研究員が自主的に研究を始めるという苦肉の策を講じ、これが図に当たった」**

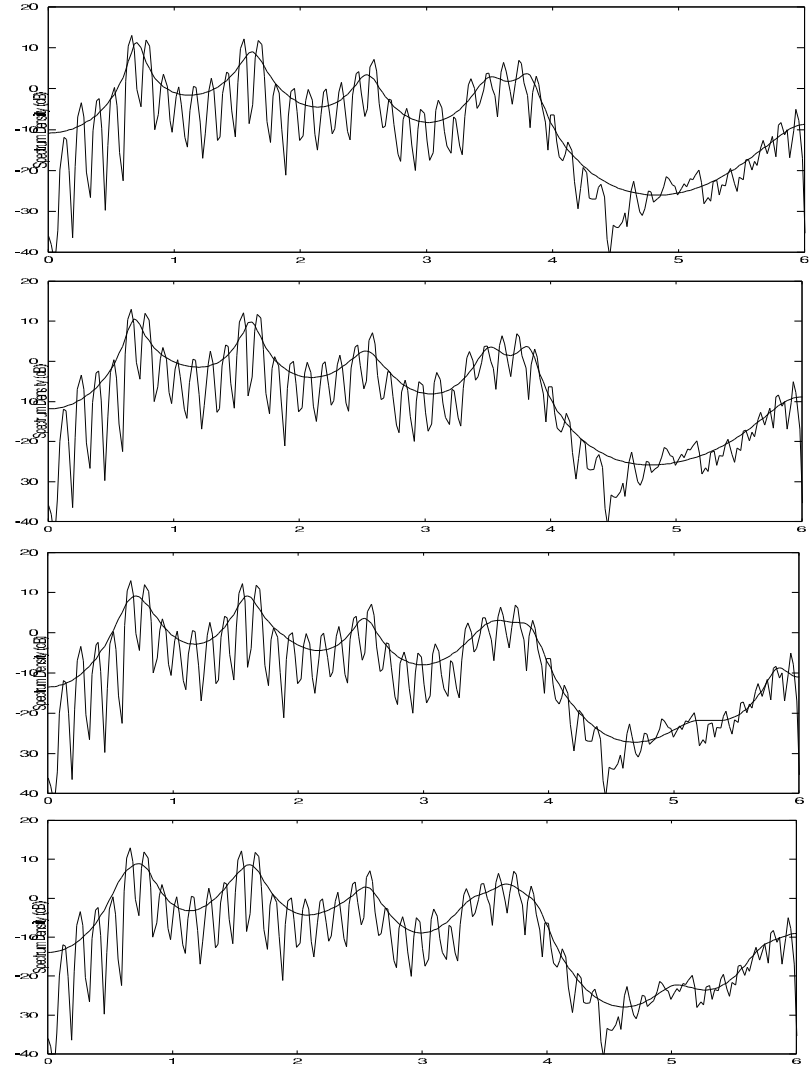
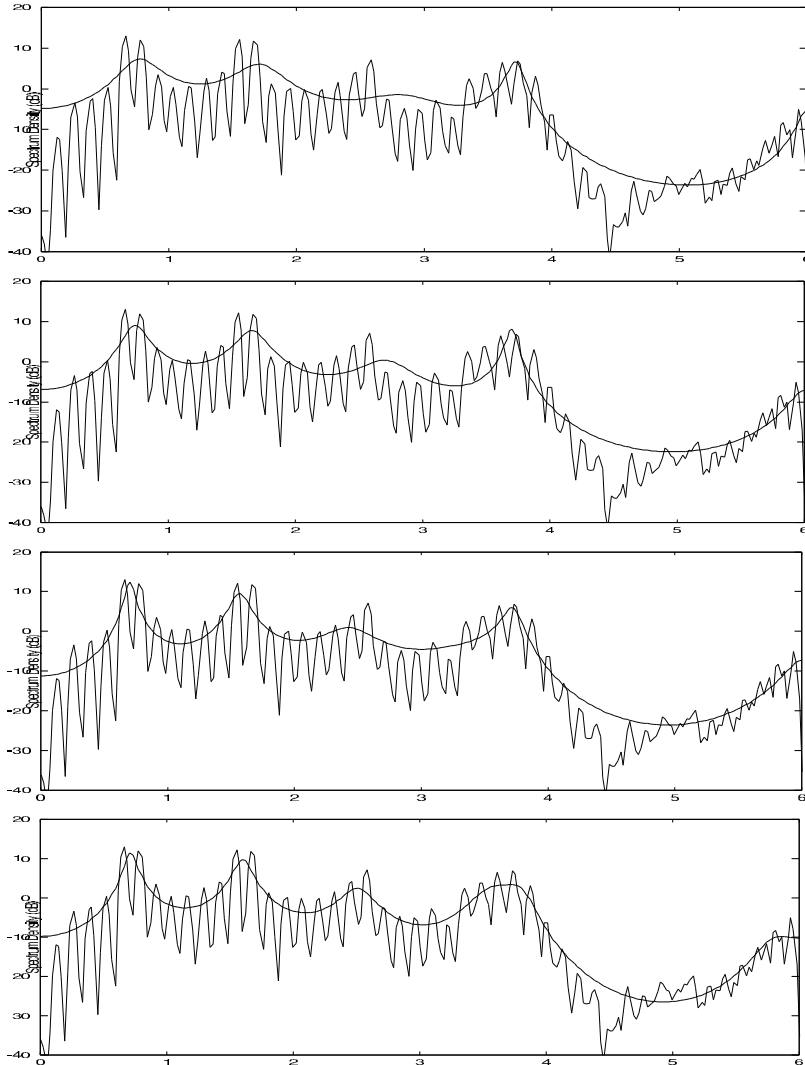
Example of speech spectrum estimation by ML estimation

■ 1~8th order



Example of speech spectrum estimation by ML estimation

■ 9, 10, 12, 14, 16, 18, 22, 26th order



Wiener Filter

Formulation in Wiener filtering

■ Problem:

- $Y(\omega)$, $X(\omega)$, $N(\omega)$: Observed, Speech, Noise signals' Fourier transformed complex spectra
- From observed signal (mixture of noise and speech)

$$Y(\omega) = X(\omega) + N(\omega)$$

we want to estimate speech parameter θ

■ Assumption:

- Speech $X(\omega)$ and noise $N(\omega)$ are independent
- Speech $X(\omega)$ follows complex normal distribution $\mathcal{N}_{\mathbb{C}}(0, \lambda_X(\omega))$
- Noise $N(\omega)$ follows complex normal distribution $\mathcal{N}_{\mathbb{C}}(0, \lambda_N(\omega))$
- Noise power spectral density $\lambda_N(\omega)$ is known in advance (e.g., estimated during noise-only duration in observation)

MMSE estimate of $X(\omega)$

■ Formulation:

- $Y(\omega) = X(\omega) + N(\omega)$
- $N(\omega) \sim \mathcal{N}_{\mathbb{C}}(N(\omega); 0, \lambda_N(\omega)), \quad X(\omega) \sim \mathcal{N}_{\mathbb{C}}(X(\omega); 0, \lambda_X(\omega))$

■ We want $\hat{X}_{\text{MMSE}}(\omega) = \mathbb{E}[X(\omega)|Y(\omega)]$

Property of multivariate Gaussian distribution

$$p(y, x) = \mathcal{N} \left(\begin{bmatrix} y \\ x \end{bmatrix}; \begin{bmatrix} \bar{y} \\ \bar{x} \end{bmatrix}, \begin{bmatrix} \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & \Sigma_{xx} \end{bmatrix} \right) \rightarrow \mathbb{E}[x|y] = \bar{x} + \Sigma_{xy} \Sigma_{yy}^+ (y - \bar{y})$$

$$p(Y(\omega), X(\omega)) = \mathcal{N}_{\mathbb{C}} \left(\begin{bmatrix} Y(\omega) \\ X(\omega) \end{bmatrix}; \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \lambda_X(\omega) + \lambda_N(\omega) & \lambda_X(\omega) \\ \lambda_X(\omega) & \lambda_X(\omega) \end{bmatrix} \right)$$

$$\rightarrow \mathbb{E}[X(\omega)|Y(\omega)] = \frac{\lambda_X(\omega)}{\lambda_X(\omega) + \lambda_N(\omega)} Y(\omega) \quad (\text{same as Wiener filter!})$$

■ Linear estimator

$$\hat{s}(t) = \int_{-\infty}^{\infty} h(t - \tau)x(\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau$$

- Model estimated signal using linear combination of observed data $x(t - \tau)$

■ MMSE criterion

$$J[h(t)] = \mathbb{E}[|\tilde{s}(t) - s(t)|^2]$$

- Here we want to get optimal $h(t)$ so that J is minimum

Solution of Wiener-Hopf equation

Review

- $h(t)$ is given by the following eq. in anti-causal case

$$R_{sx}(\tau) = \int_{-\infty}^{\infty} R_{xx}(\tau - \alpha)h(\alpha)d\alpha$$

Applying Fourier transform, we get

$$S_{sx}(\omega) = S_{xx}(\omega)H(\omega) \longrightarrow H(\omega) = \frac{S_{sx}(\omega)}{S_{xx}(\omega)}$$

- In case of $R_{sn}(\tau) = 0$

$$R_{sx}(\tau) = R_{ss}(\tau), \quad R_{xx}(\tau) = R_{ss}(\tau) + R_{nn}(\tau)$$

$$H(\omega) = \frac{S_{ss}(\omega)}{S_{ss}(\omega) + S_{nn}(\omega)}$$

Anti-causal Wiener filter

Cf. result using property of
multivariate Gaussian distribution

Kalman Filter

Discrete-time Kalman filter's model

System model: $x_k = \Phi_{k,k-1}x_{k-1} + \underline{B_k v_k}$

Process noise

Observation model: $y_k = C_k x_k + \underline{w_k}$

Observation noise

k : time index

■ Assumption

- v_k, w_k are mutually independent Gaussian noises

$$\mathbb{E}[v_k] = 0 \quad \mathbb{E}[v_k v_n^T] = V_k \delta_{kn}$$

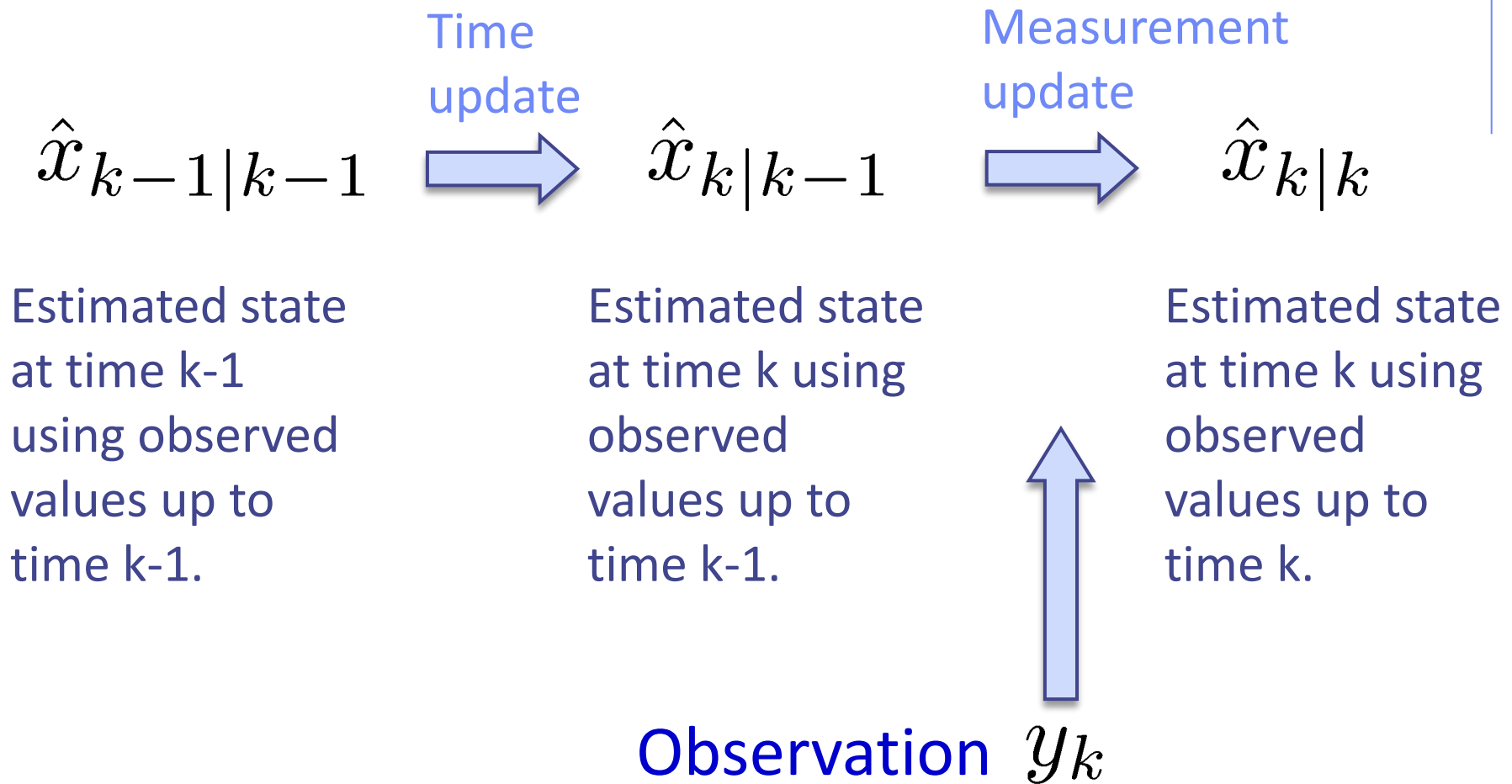
$$\mathbb{E}[w_k] = 0 \quad \mathbb{E}[w_k w_n^T] = W_k \delta_{kn}$$

$$\mathbb{E}[v_k w_n^T] = 0$$

- Parameter: $\Phi_{k,k-1}, B_k, C_k$ and V_k, W_k are known

Process flow of discrete-time Kalman filter

State estimate



Kalman filter algorithm overview

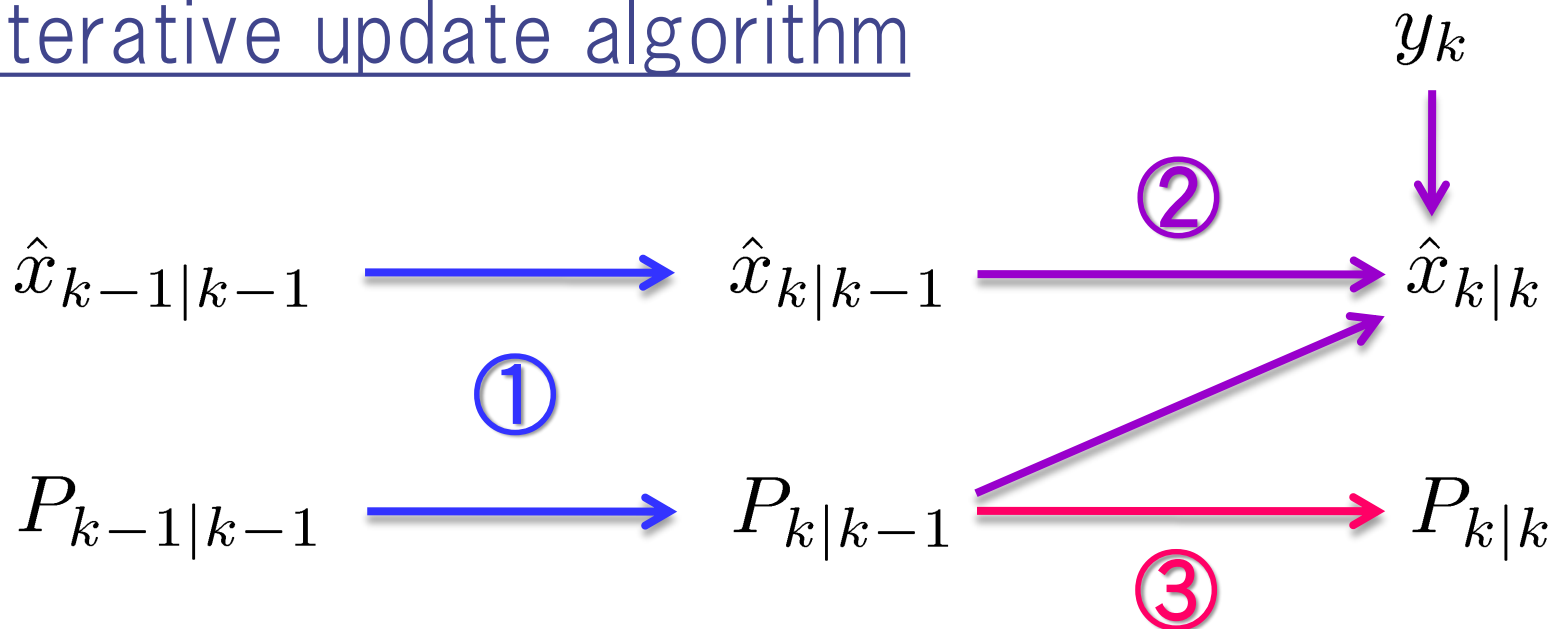
■ Algorithm purpose

■ $\hat{x}_{k|k}, P_{k|k}$ are updated using y_k & $\hat{x}_{k-1|k-1}, P_{k-1|k-1}$

■ where

$$P_{n|m} = \text{cov}(x_n - \hat{x}_{n|m})$$

Iterative update algorithm



① Time update for state See 信号处理論2講義資料 for details

■ $\hat{x}_{k-1|k-1} \longrightarrow \hat{x}_{k|k-1}$

$$\hat{x}_{k|k-1} = \Phi_{k,k-1} \hat{x}_{k-1|k-1}$$

■ $P_{k-1|k-1} \longrightarrow P_{k|k-1}$

$$\begin{aligned} P_{k|k-1} &= \text{cov}(x_k - \hat{x}_{k|k-1}) \\ &= \mathbb{E}[(x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^\top] \\ &= \Phi_{k,k-1} P_{k-1|k-1} \Phi_{k,k-1}^\top + \underline{B_k V_k B_k^\top} \end{aligned}$$

Increase of variance in time update

② Measurement update for state

■ $\hat{x}_{k|k-1}, y_k \longrightarrow \hat{x}_{k|k}$

■ Innovation (Error in prediction of observation)

$$\mu_k = y_k - C_k \hat{x}_{k|k-1}$$



■ Update rule

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + \underline{K}_k \mu_k$$

Kalman gain (unknown)

(Ref.) If (θ, y) obey Gaussian distribution, then

$$\theta_{\text{MMSE}} = \mathbb{E}[\theta|y] = \bar{\theta} + A_0(y - \bar{y})$$

③ Error covariance update for estimate

$$\blacksquare P_{k|k-1} \longrightarrow P_{k|k}$$

$$P_{k|k}$$

$$= \text{cov}(x_k - \hat{x}_{k|k})$$

$$= \text{cov}(x_k - (\hat{x}_{k|k-1} + K_k \underline{\mu_k}))$$

$$= \text{cov}(x_k - (\hat{x}_{k|k-1} + K_k (\underline{y_k} - C_k \hat{x}_{k|k-1})))$$

$$= \text{cov}(\underline{x_k} - (\underline{\hat{x}_{k|k-1}} + K_k (C_k \underline{x_k} + w_k - C_k \underline{\hat{x}_{k|k-1}})))$$

$$= \text{cov}((I - K_k C_k)(x_k - \hat{x}_{k|k-1}) - K_k w_k) \quad \because w \text{ and } x, \hat{x}$$

$$= \text{cov}((I - K_k C_k)(x_k - \hat{x}_{k|k-1})) + \text{cov}(K_k w_k) \quad \text{are independent}$$

$$= (I - K_k C_k) \text{cov}(x_k - \hat{x}_{k|k-1}) (I - K_k C_k)^\top + K_k \text{cov}(w_k) K_k^\top$$

$$= (I - K_k C_k) P_{k|k-1} (I - K_k C_k)^\top + K_k W_k K_k^\top$$

② Optimal Kalman gain

■ Optimization problem for Kalman gain

$$\mathbb{E}[\|x_k - \hat{x}_{k|k}\|_2^2] \rightarrow \text{minimize}$$

$$\begin{aligned}\mathbb{E}[\|x_k - \hat{x}_{k|k}\|_2^2] &= \mathbb{E}[(x_k - \hat{x}_{k|k})^\top (x_k - \hat{x}_{k|k})] \\ &= \text{tr}(\mathbb{E}[(x_k - \hat{x}_{k|k})(x_k - \hat{x}_{k|k})^\top]) \\ &= \text{tr}(\underline{P_{k|k}}) \\ &= \text{tr}((I - K_k C_k) P_{k|k-1} (I - K_k C_k)^\top + K_k W_k K_k^\top)\end{aligned}$$

② Optimal Kalman gain

■ Optimization criteria

$$\begin{aligned}\text{tr}(P_{k|k}) &= \text{tr}\left((I - K_k C_k) P_{k|k-1} (I - K_k C_k)^\top + K_k W_k K_k^\top\right) \\ &= \text{tr}\left(P_{k|k-1} - \underline{K_k} C_k P_{k|k-1} - P_{k|k-1} C_k^\top \underline{K_k}^\top \right. \\ &\quad \left. + \underline{K_k} C_k P_{k|k-1} C_k^\top \underline{K_k}^\top + \underline{K_k} W_k \underline{K_k}^\top\right)\end{aligned}$$

$$\blacksquare \text{tr}(P_{k|k}) \text{ is minimized} \rightarrow \frac{\partial \text{tr}(P_{k|k})}{\partial K_k} = 0$$

$$\begin{aligned}\frac{\partial \text{tr}(P_{k|k})}{\partial K_k} &= -2P_{k|k-1} C_k^\top + 2K_k (C_k P_{k|k-1} C_k^\top + W_k) \\ &= 0\end{aligned}$$

$\therefore$ Trace's differentiation formula

$$\frac{\partial \text{tr}(MK^\top)}{\partial K} = M, \quad \frac{\partial \text{tr}(KMK^\top)}{\partial K} = KM^\top + KM$$

$$\therefore \hat{K}_k = P_{k|k-1} C_k^\top (C_k P_{k|k-1} C_k^\top + W_k)^{-1}$$

③ Error covariance under optimal Kalman gain

- $\hat{K}_k$ is substituted into $P_{k|k}$

$$P_{k|k} = (I - K_k C_k) P_{k|k-1} (I - K_k C_k)^T + K_k W_k K_k^T$$

$$\hat{K}_k = P_{k|k-1} C_k^T (C_k P_{k|k-1} C_k^T + W_k)^{-1}$$

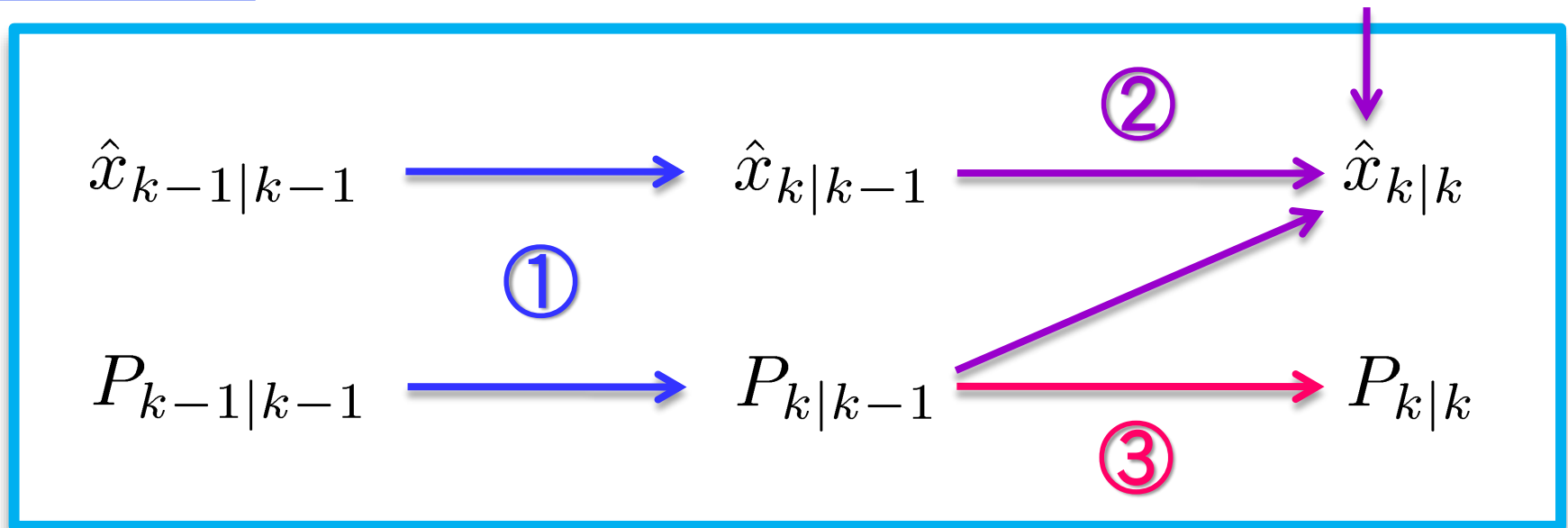
(Optimal Kalman gain)



$$\hat{K}_k (C_k P_{k|k-1} C_k^T + W_k) \hat{K}_k^T = P_{k|k-1} C_k^T \hat{K}_k^T$$


$$P_{k|k} = (I - \hat{K}_k C_k) P_{k|k-1}$$

Summary of Kalman filter algorithm



- ① $\hat{x}_{k|k-1} = \Phi_{k,k-1} \hat{x}_{k-1|k-1}$
 $P_{k|k-1} = \Phi_{k,k-1} P_{k-1|k-1} \Phi_{k,k-1}^T + B_k V_k B_k^T$
- ② $\hat{x}_{k|k} = \hat{x}_{k|k-1} + P_{k|k-1} C_k^T (C_k P_{k|k-1} C_k^T + W_k)^{-1} (y_k - C_k \hat{x}_{k|k-1})$
- ③ $P_{k|k} = (I - P_{k|k-1} C_k^T (C_k P_{k|k-1} C_k^T + W_k)^{-1} C_k) P_{k|k-1}$

Probability distribution in iterative update

- $p(x_{k-1} | y_{1:k-1})$

- Given the observation signals from time t_1 to t_{k-1} , the posterior probability distribution of the state estimate at time t_{k-1} .

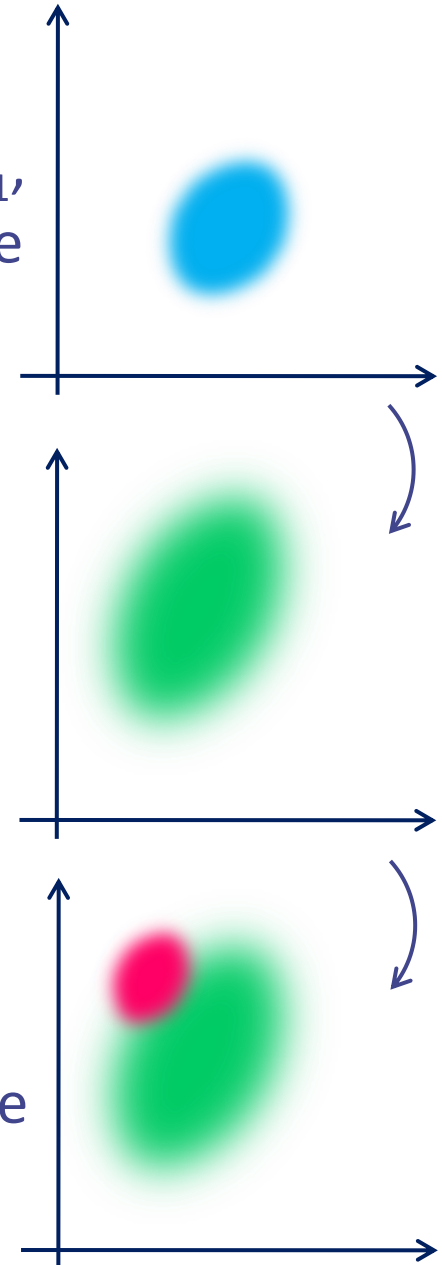
$\hat{x}_{k-1|k-1}, P_{k-1|k-1}$: mean & covariance

- $p(x_k | y_{1:k-1})$

- The prior probability distribution of the state estimate at time t_k , inferred from the above posterior distribution at time t_{k-1} .

- $p(x_k | y_{1:k}) = p(x_k | y_k, y_{1:k-1})$

- The prior probability distribution of the state estimate at time t_k , obtained based on the above prior distribution and observed signal at time t_k .



Bayesian interpretation of Kalman filter

$$p(x_{k-1}|y_{1:k-1}) = \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$$

From system model:

$$p(x_k|x_{k-1}) = \mathcal{N}(\Phi_{k,k-1}x_{k-1}, B_k V_k B_k^\top)$$

From observation model:

$$p(y_k|x_k) = \mathcal{N}(C_k x_k, W_k)$$

Time
update

$$p(x_k|y_{1:k-1}) = \int p(x_k|x_{k-1}) p(x_{k-1}|y_{1:k-1}) dx_{k-1}$$

Measurement
update

$$\begin{aligned} p(x_k|y_{1:k}) &= \frac{p(y_k|x_k) p(x_k|y_{1:k-1})}{p(y_k|y_{1:k-1})} \\ &= \frac{p(y_k|x_k) p(x_k|y_{1:k-1})}{\int p(y_k|x_k) p(x_k|y_{1:k-1}) dx_k} \end{aligned}$$

$$p(x_k|y_{1:k}) = \mathcal{N}(\hat{x}_{k|k}, P_{k|k})$$